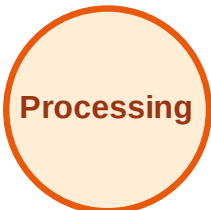


BFS Traversal

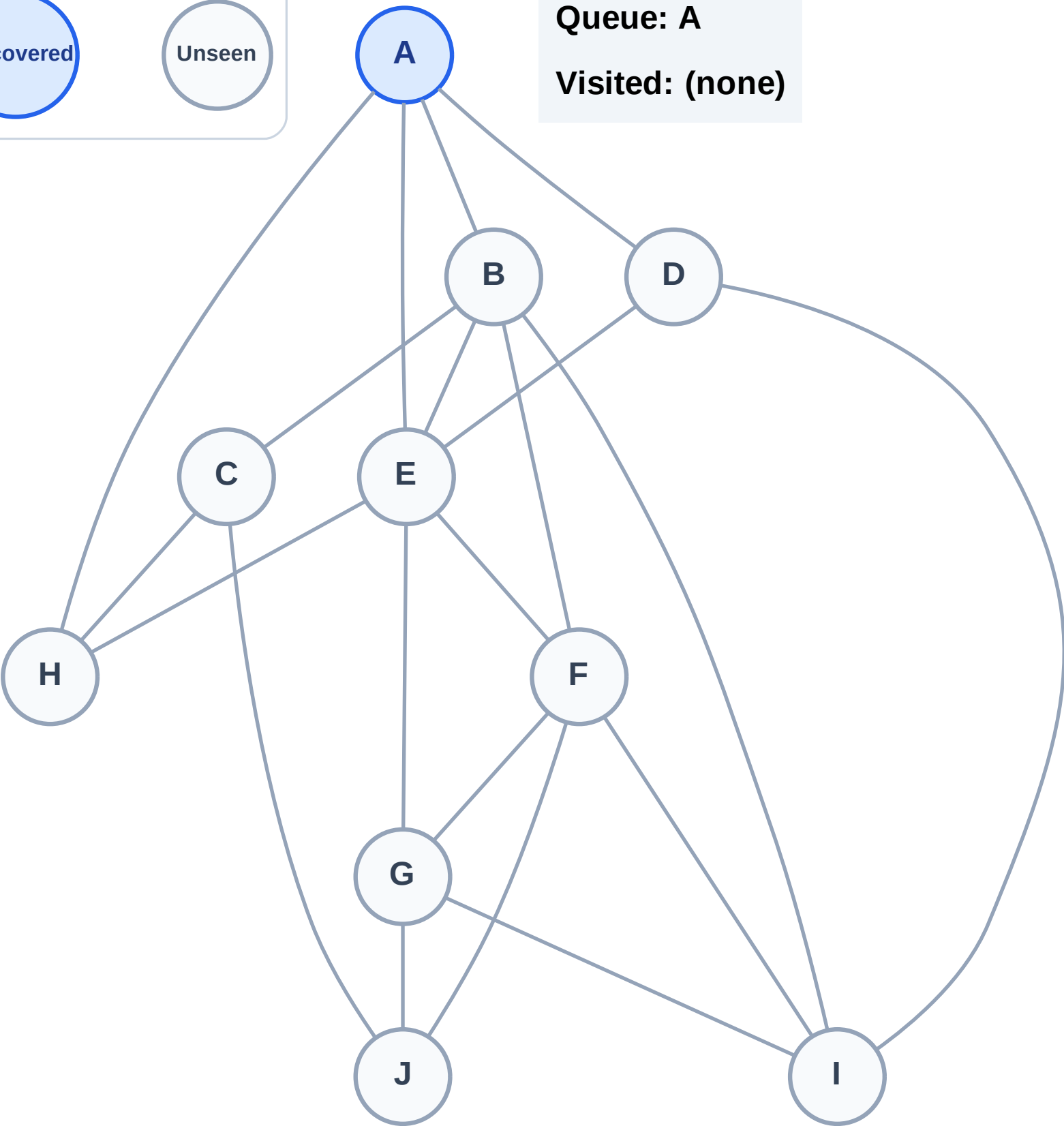
Step 1: Start at A

Legend



Queue: A

Visited: (none)



BFS Traversal

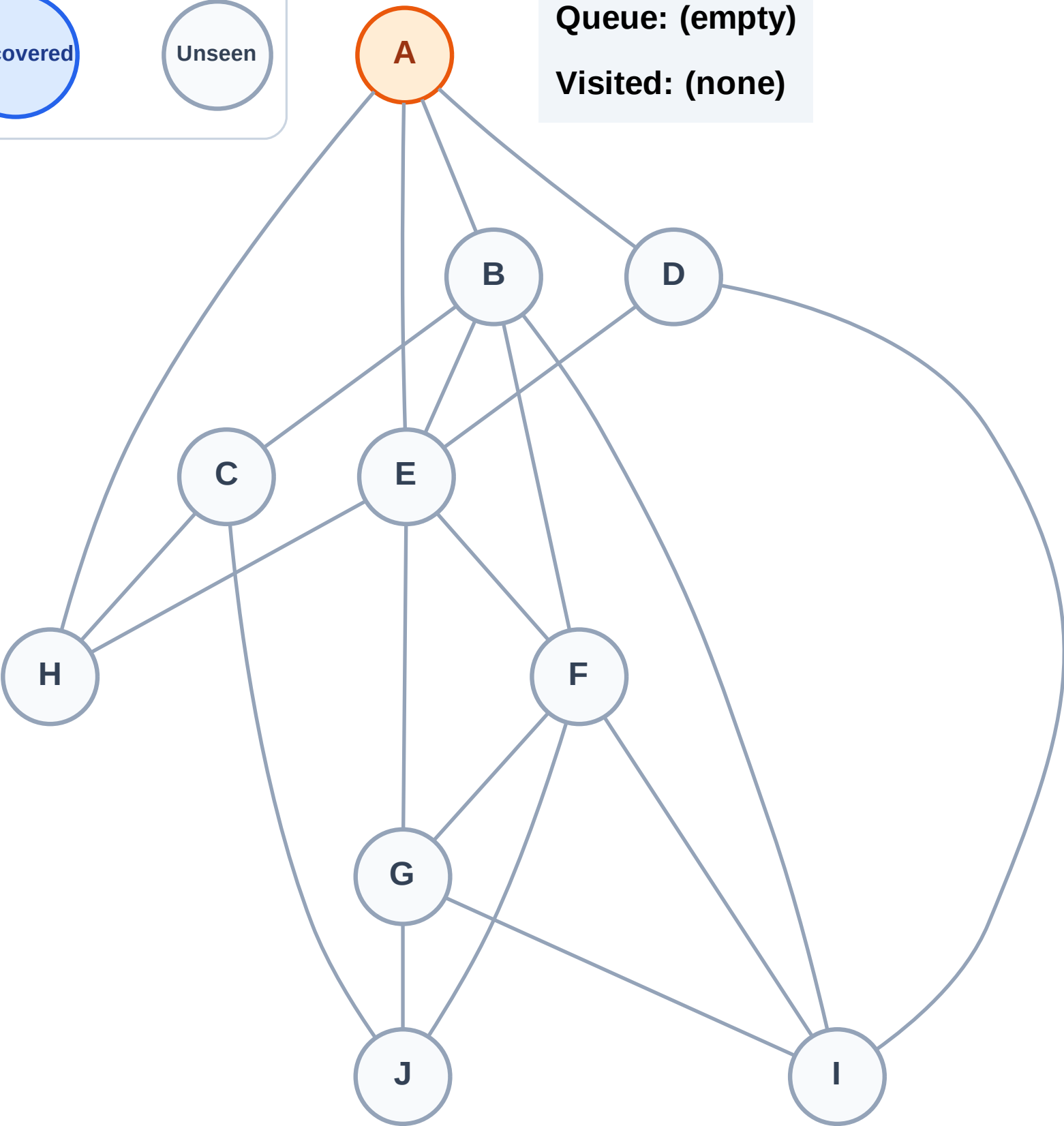
Step 2: Process node A

Legend



Queue: (empty)

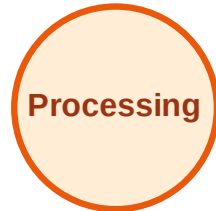
Visited: (none)



BFS Traversal

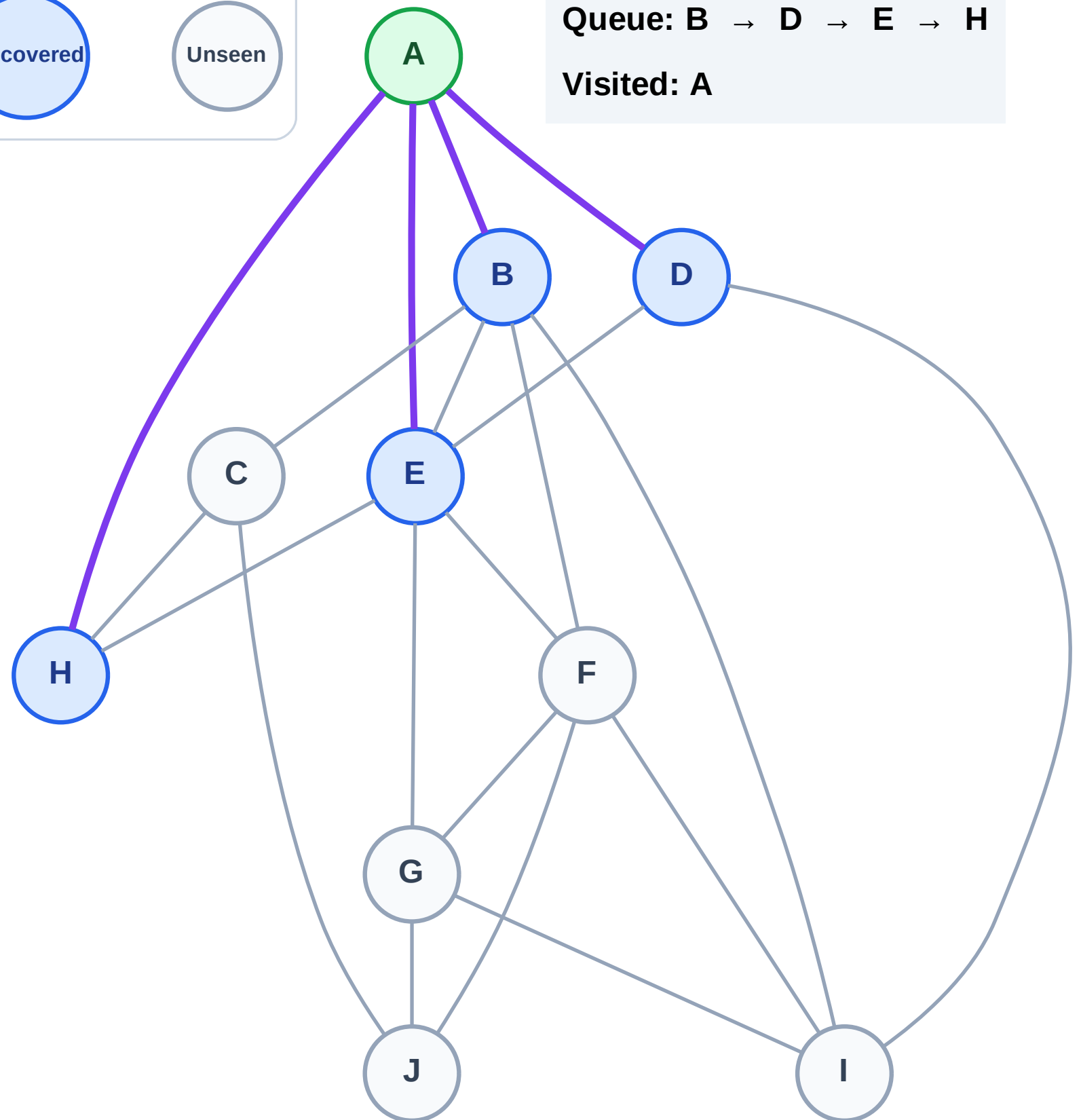
Step 3: Discover B, D, E, H from A

Legend



Queue: B → D → E → H

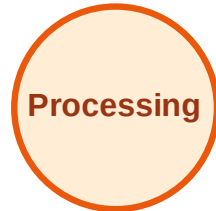
Visited: A



BFS Traversal

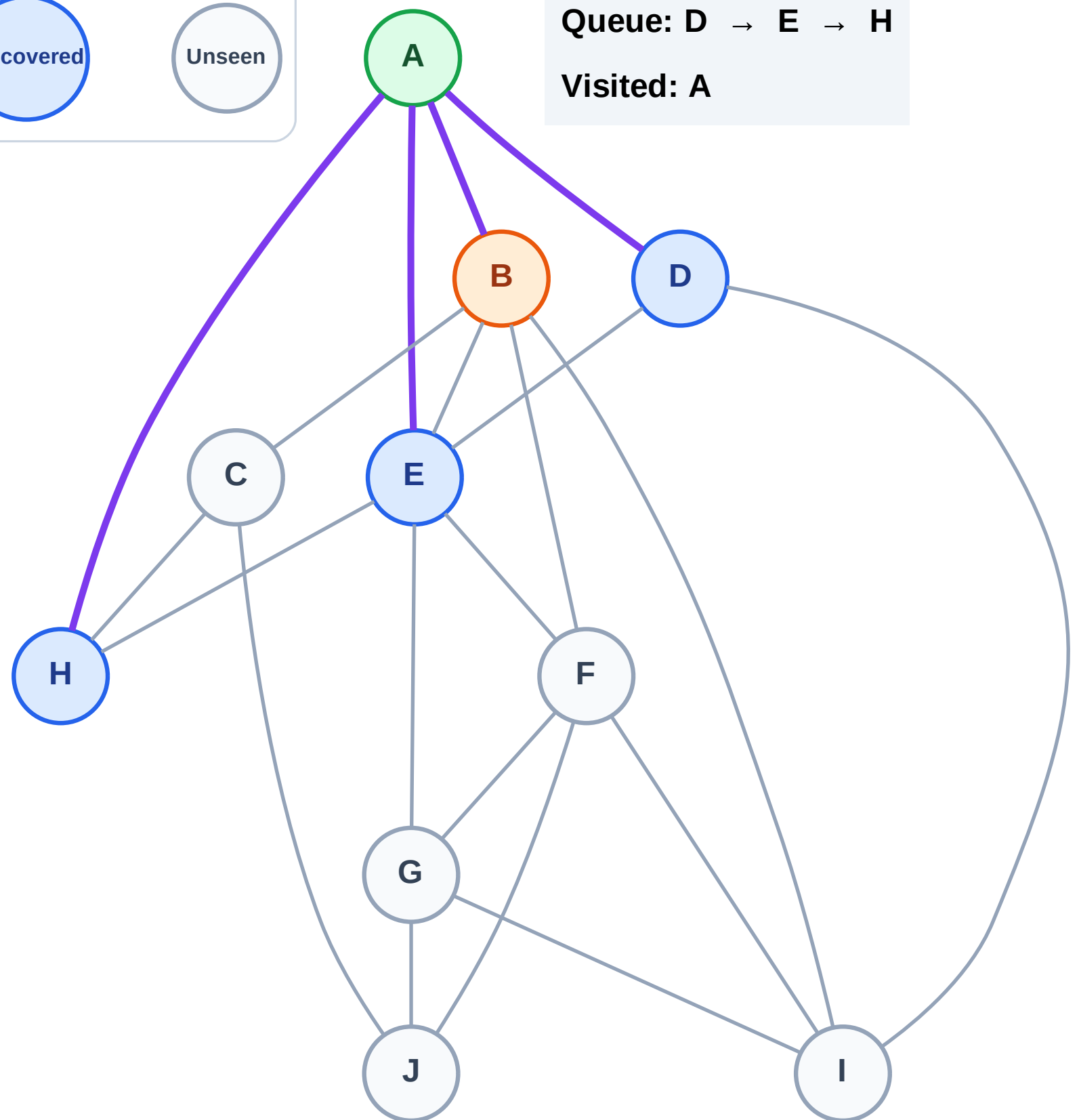
Step 4: Process node B

Legend



Queue: D → E → H

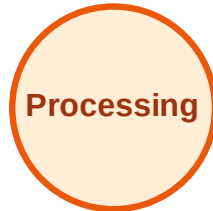
Visited: A



BFS Traversal

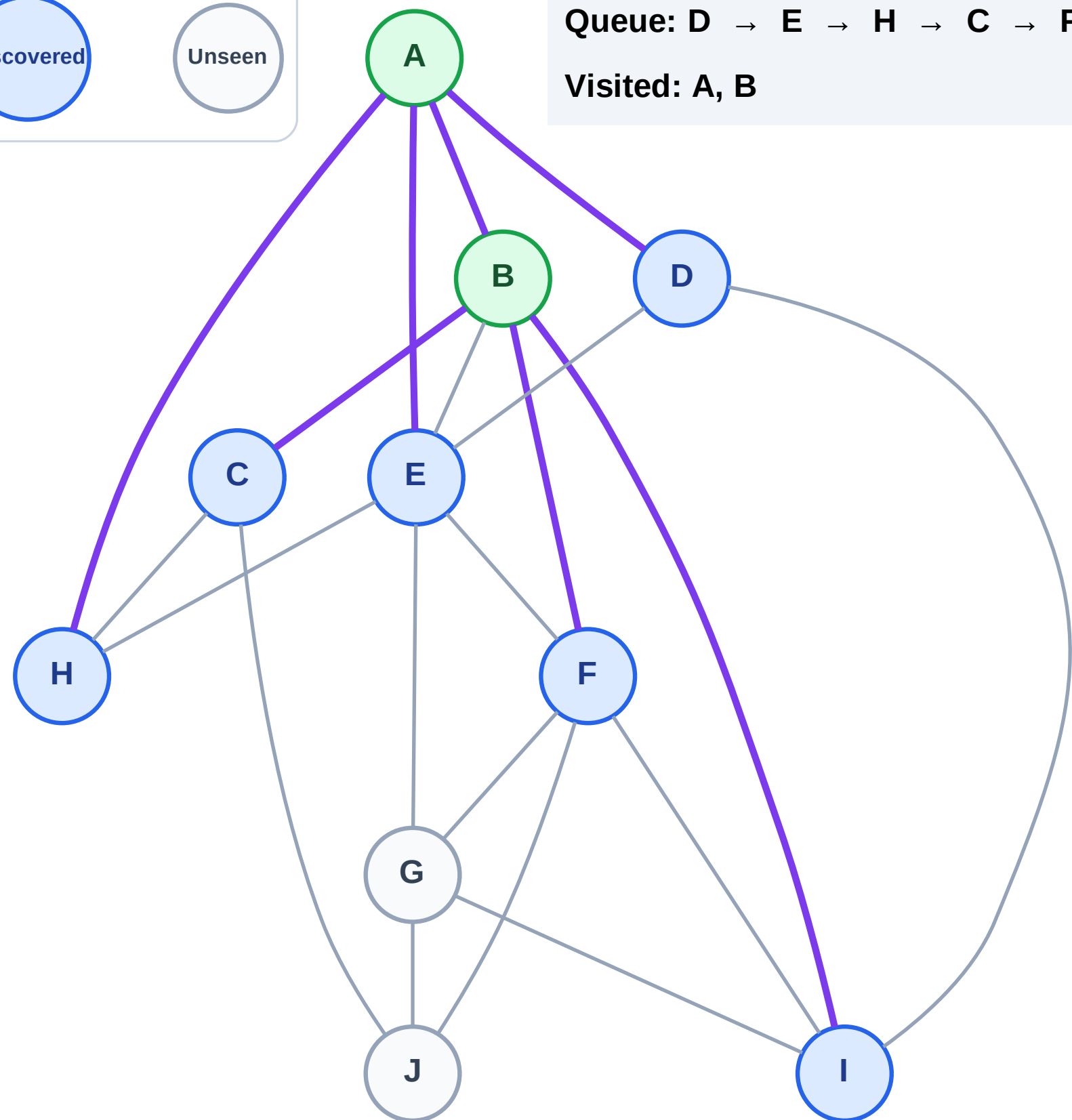
Step 5: Discover C, F, I from B

Legend



Queue: D → E → H → C → F → I

Visited: A, B



BFS Traversal

Step 6: Process node D

Legend

Complete

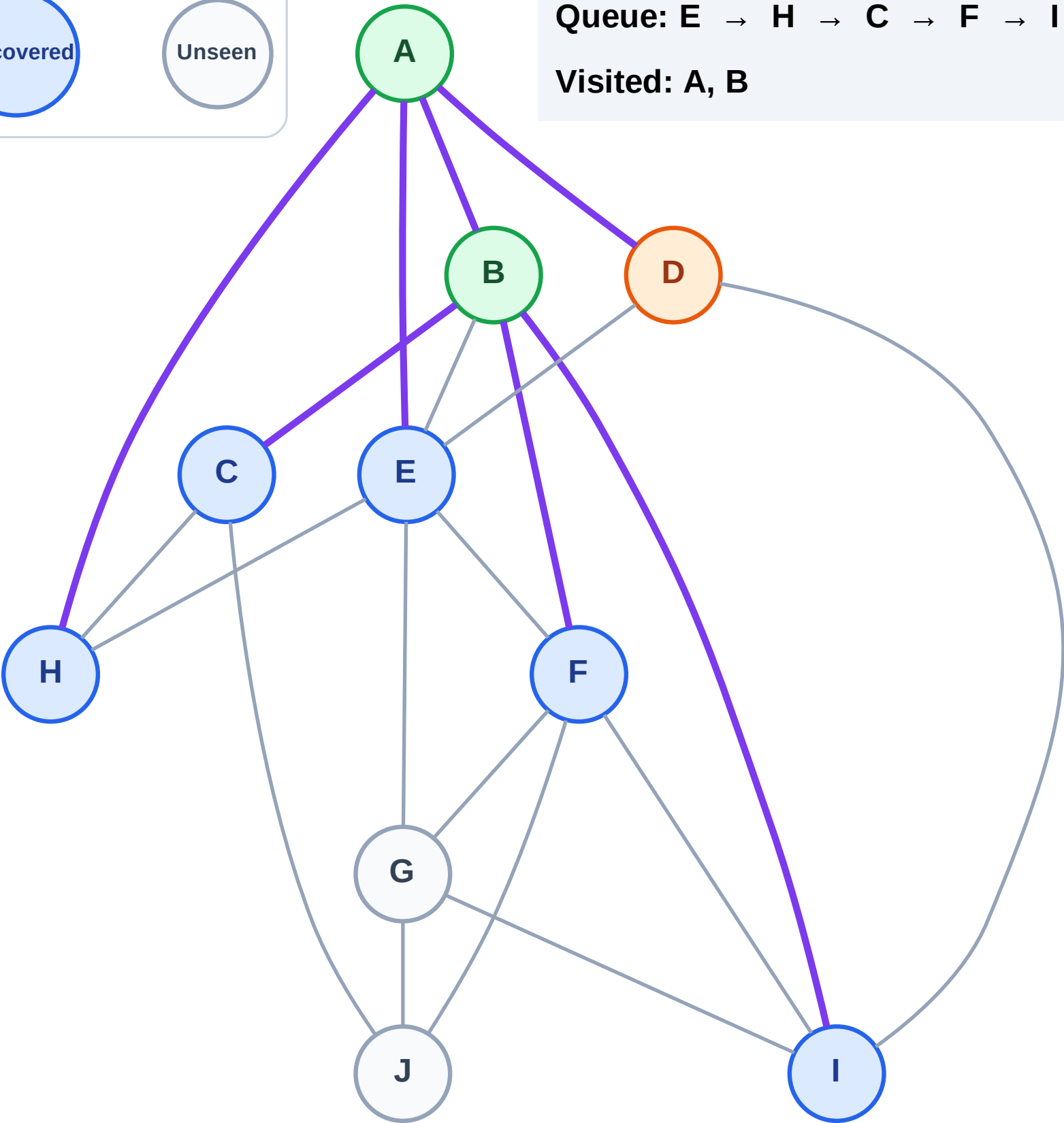
Processing

Discovered

Unseen

Queue: E → H → C → F → I

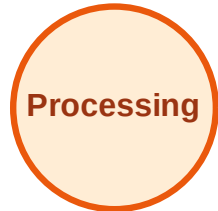
Visited: A, B



BFS Traversal

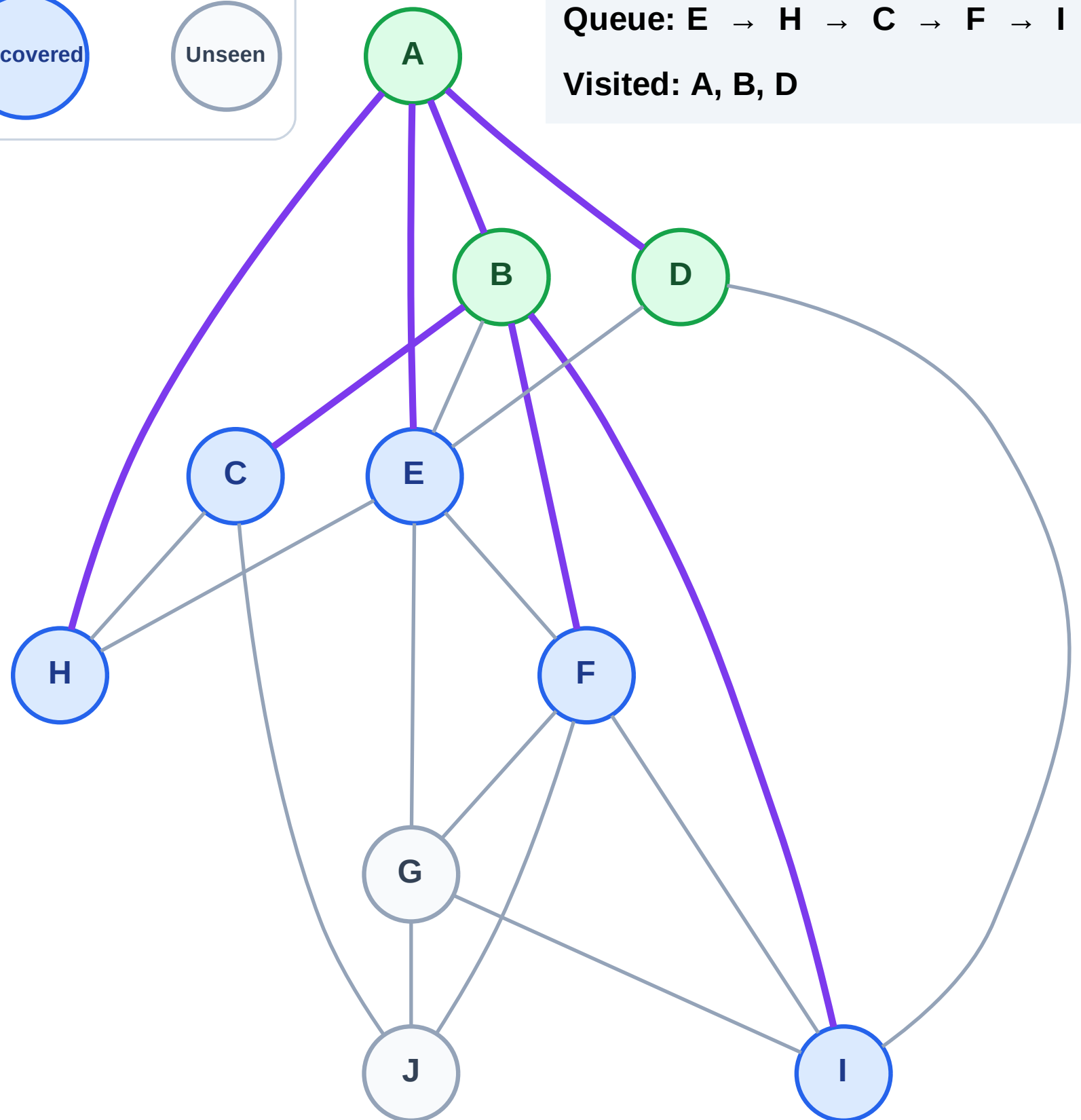
Step 7: No new neighbors from D

Legend



Queue: E → H → C → F → I

Visited: A, B, D



BFS Traversal

Step 8: Process node E

Legend

Complete

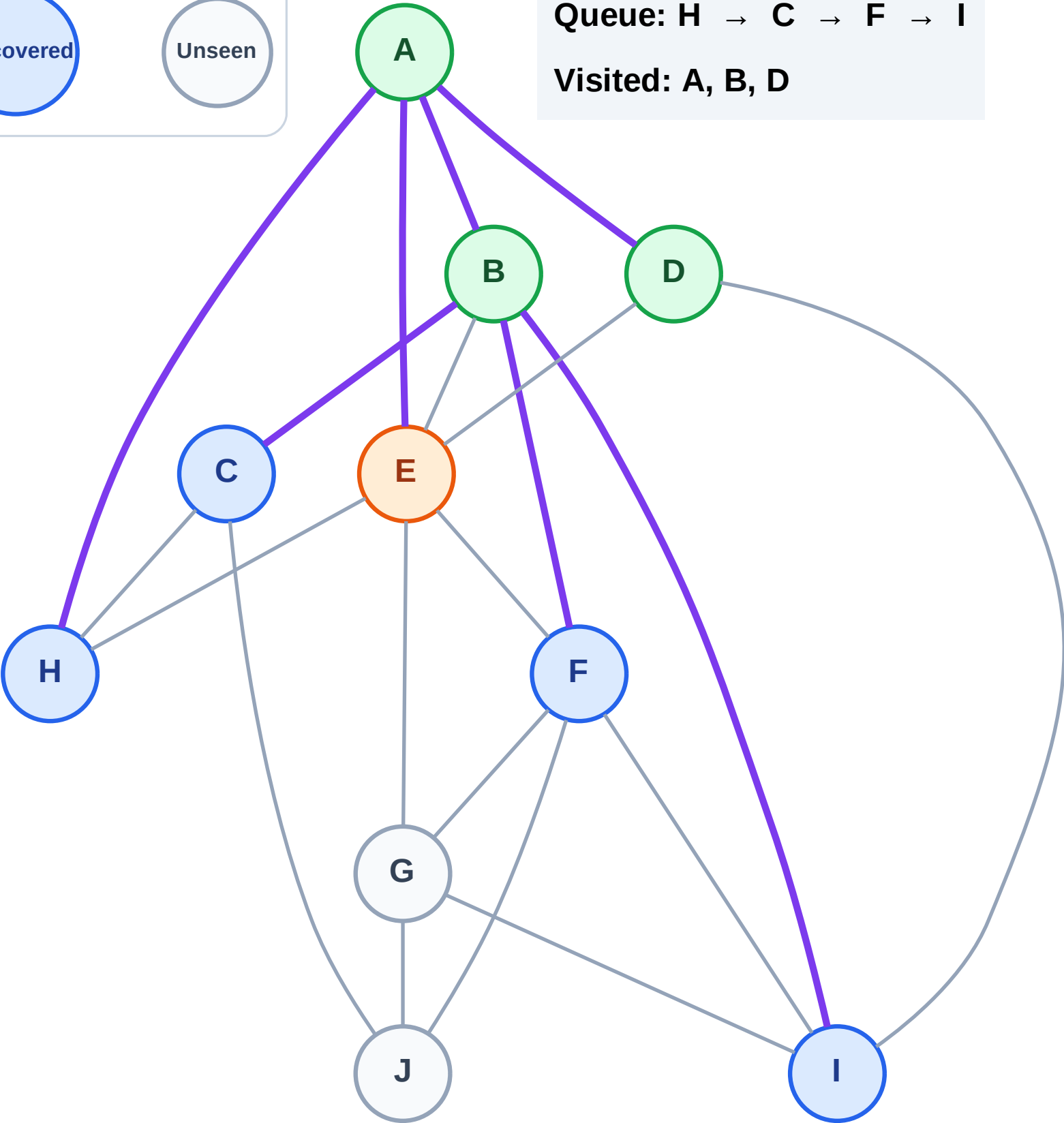
Processing

Discovered

Unseen

Queue: H → C → F → I

Visited: A, B, D



BFS Traversal

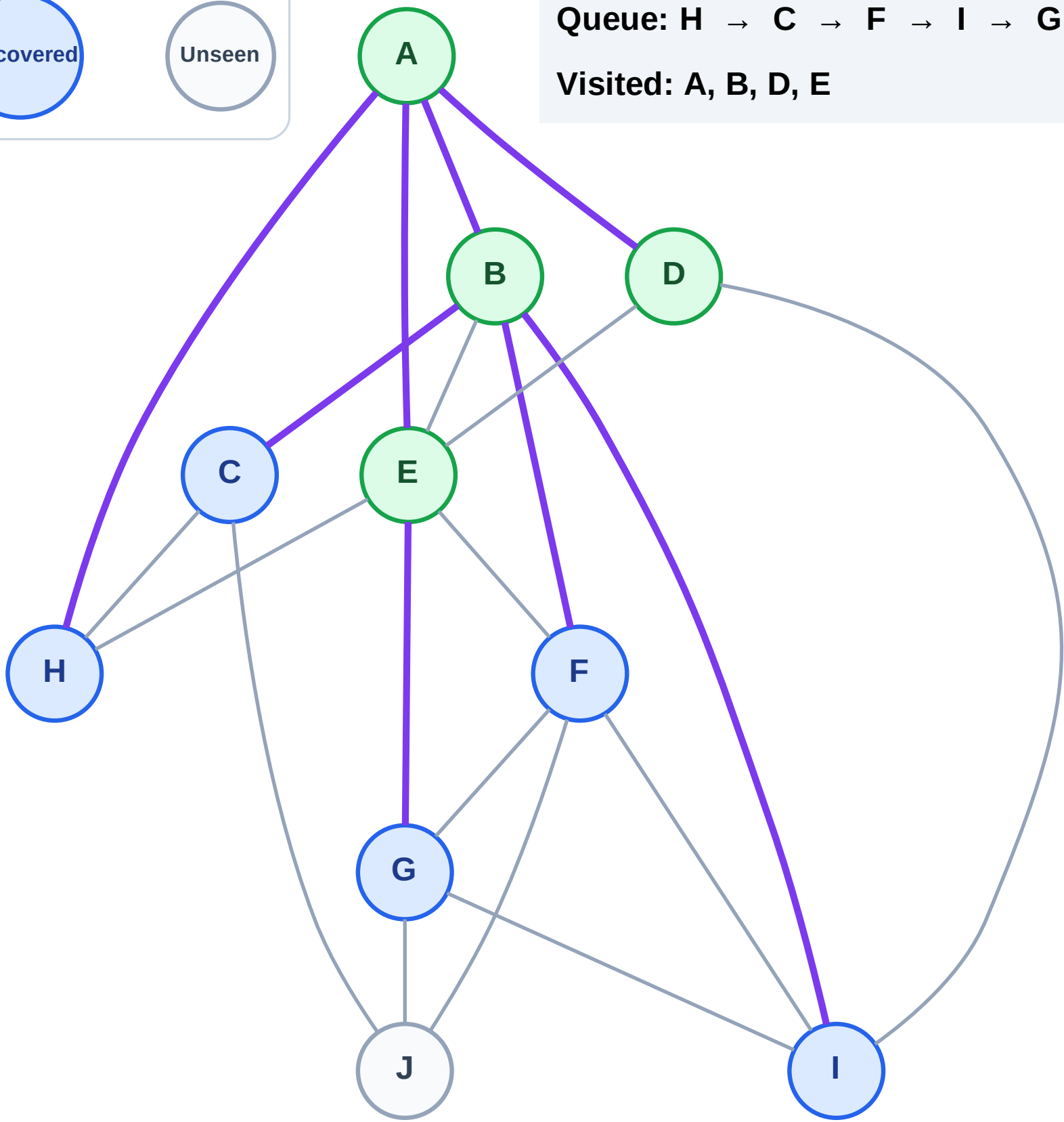
Step 9: Discover G from E

Legend



Queue: H → C → F → I → G

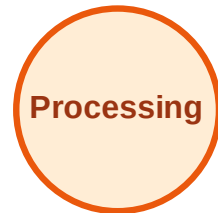
Visited: A, B, D, E



BFS Traversal

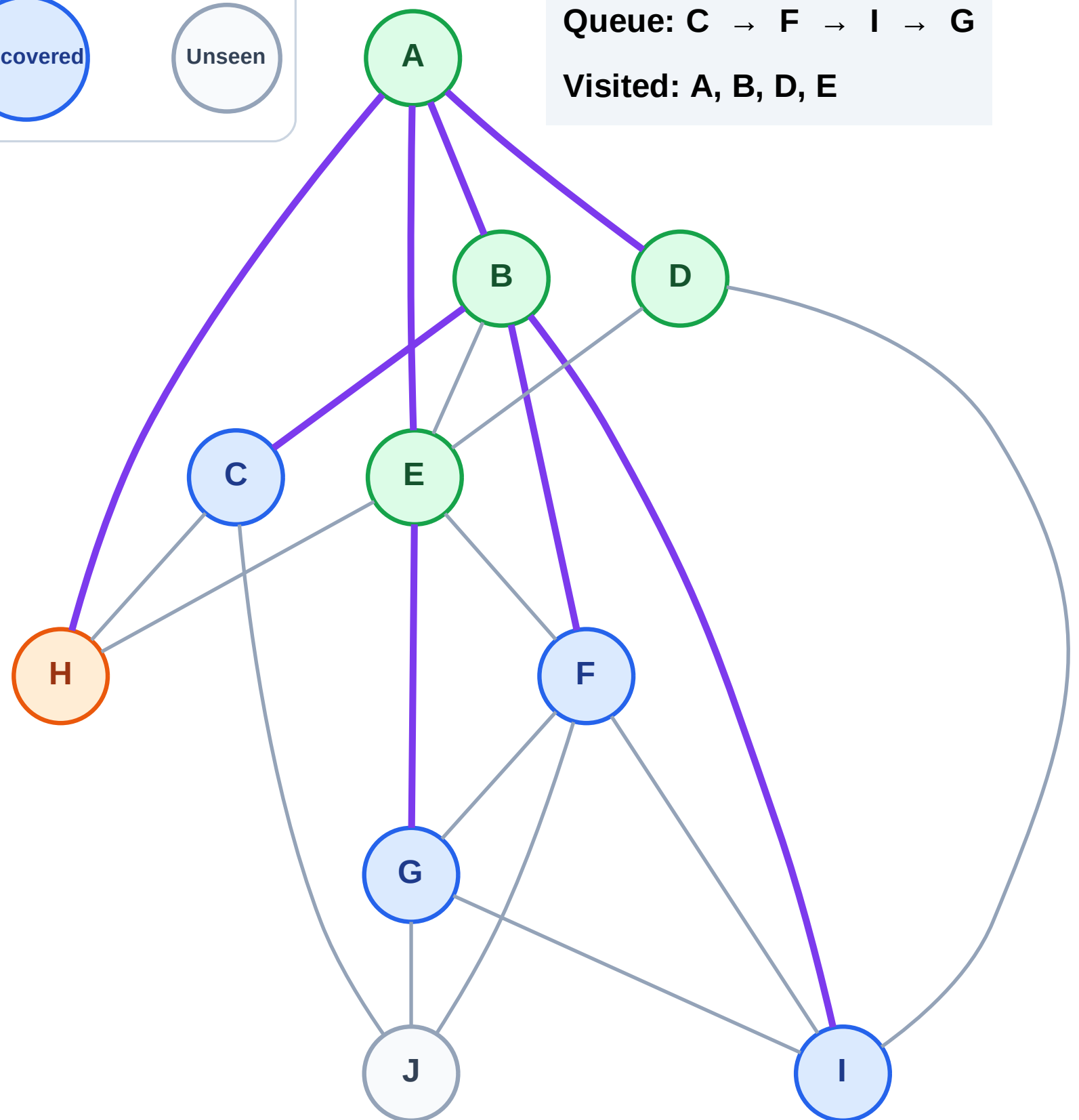
Step 10: Process node H

Legend



Queue: C → F → I → G

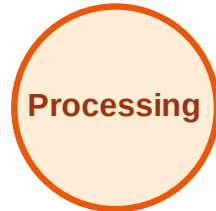
Visited: A, B, D, E



BFS Traversal

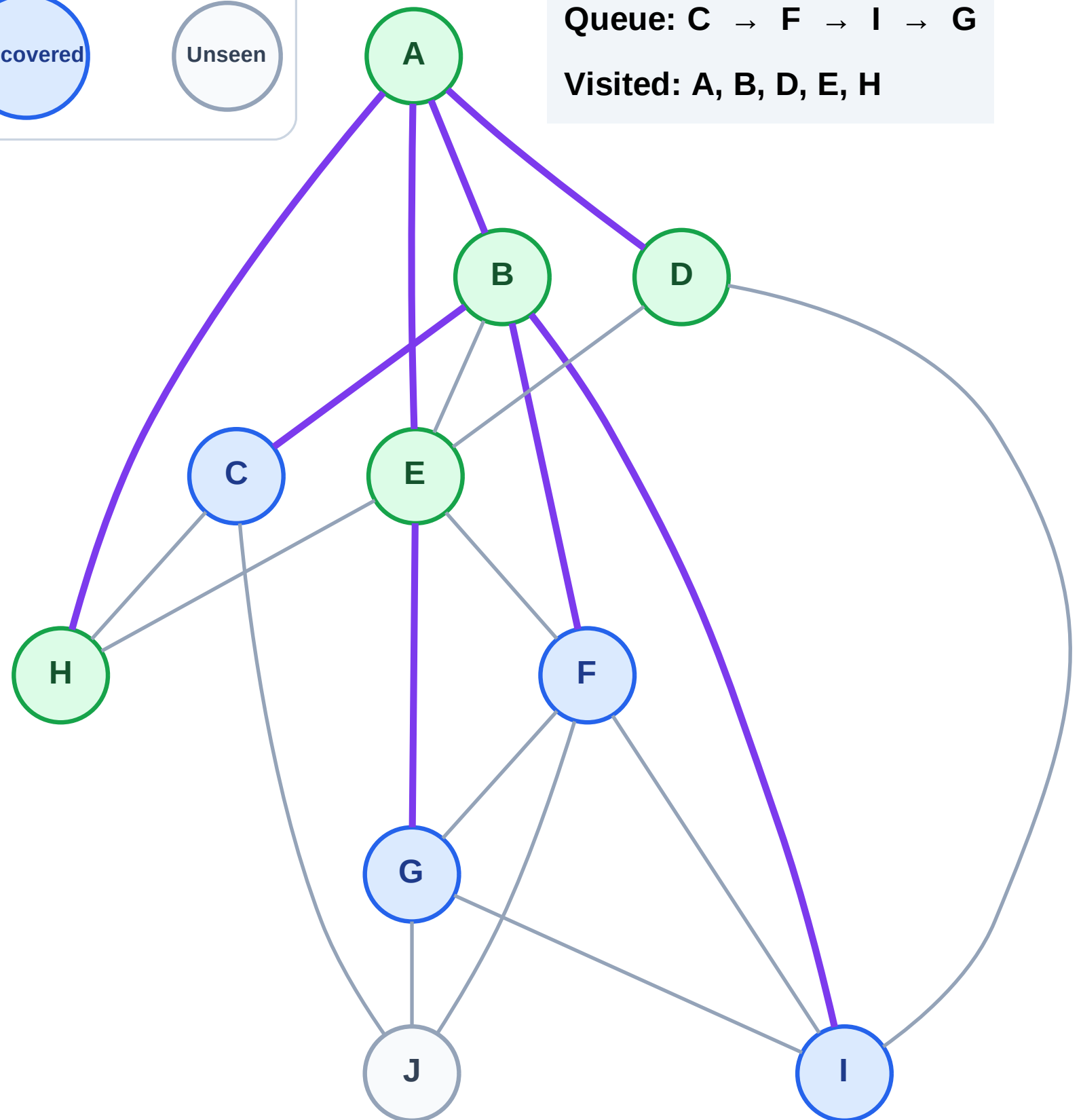
Step 11: No new neighbors from H

Legend



Queue: C → F → I → G

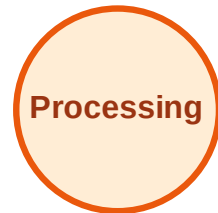
Visited: A, B, D, E, H



BFS Traversal

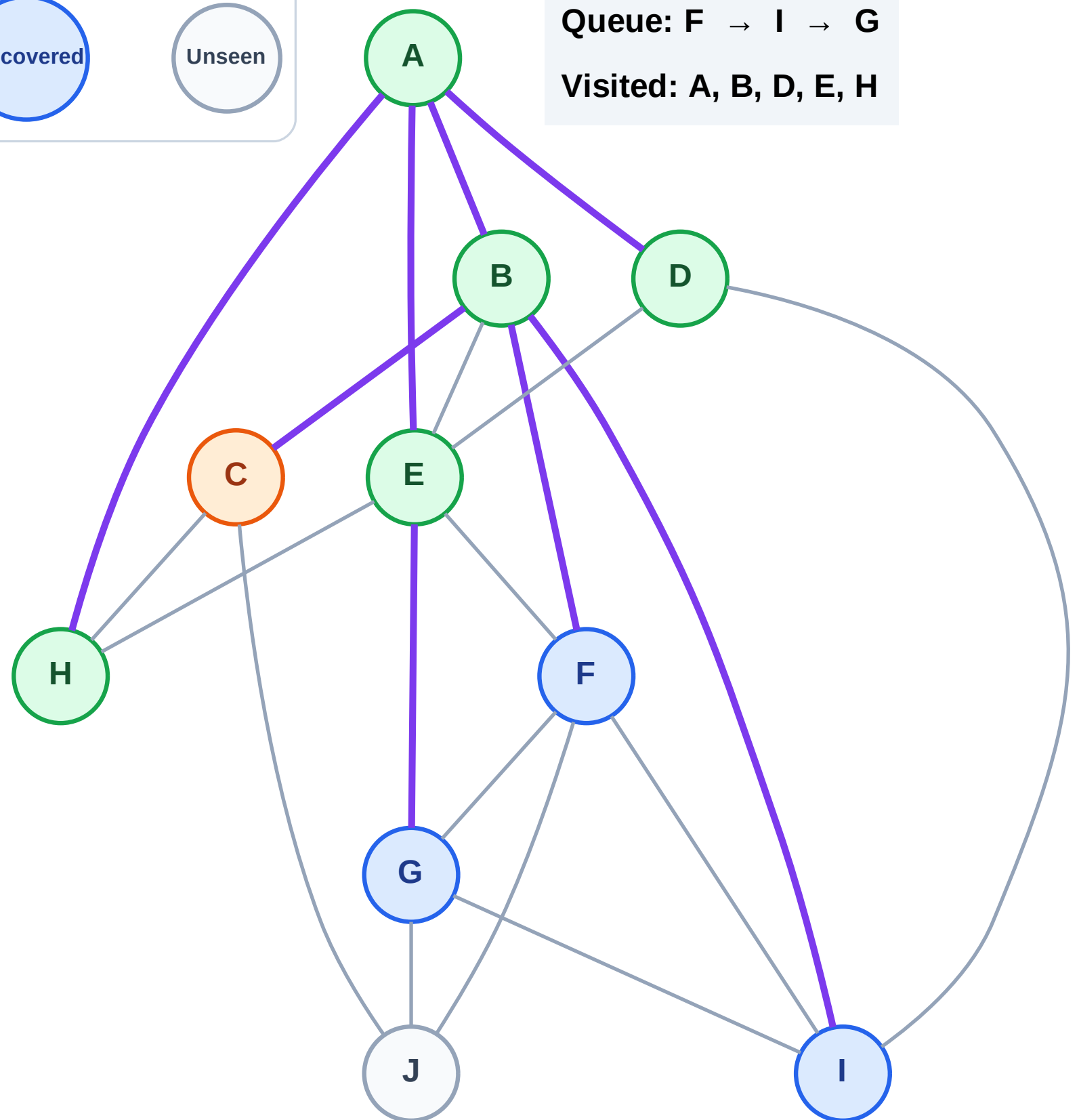
Step 12: Process node C

Legend



Queue: F → I → G

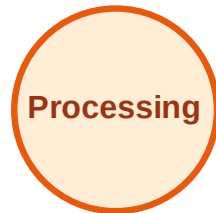
Visited: A, B, D, E, H



BFS Traversal

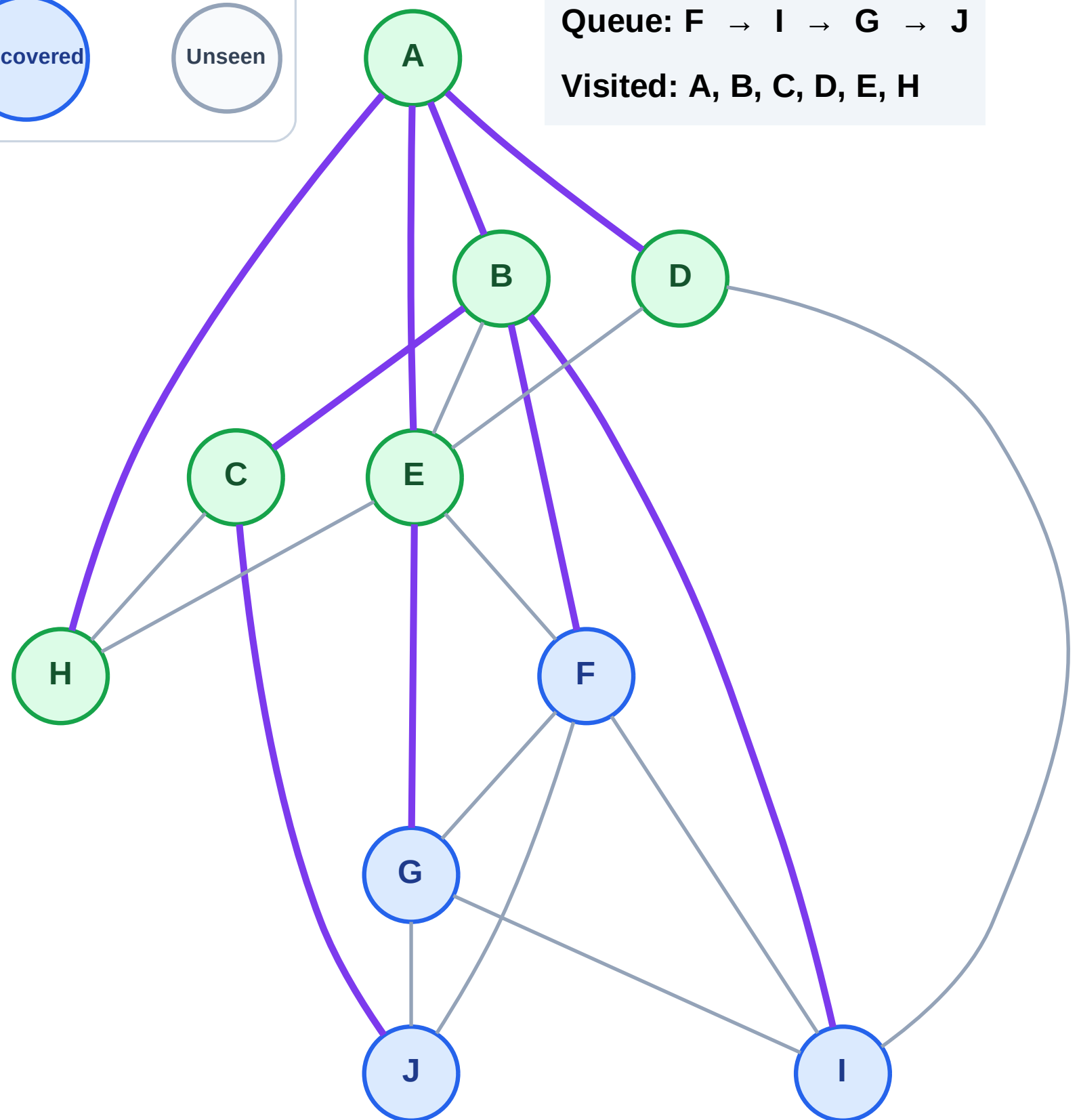
Step 13: Discover J from C

Legend



Queue: F → I → G → J

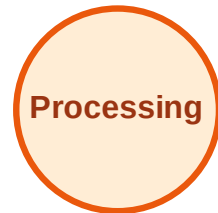
Visited: A, B, C, D, E, H



BFS Traversal

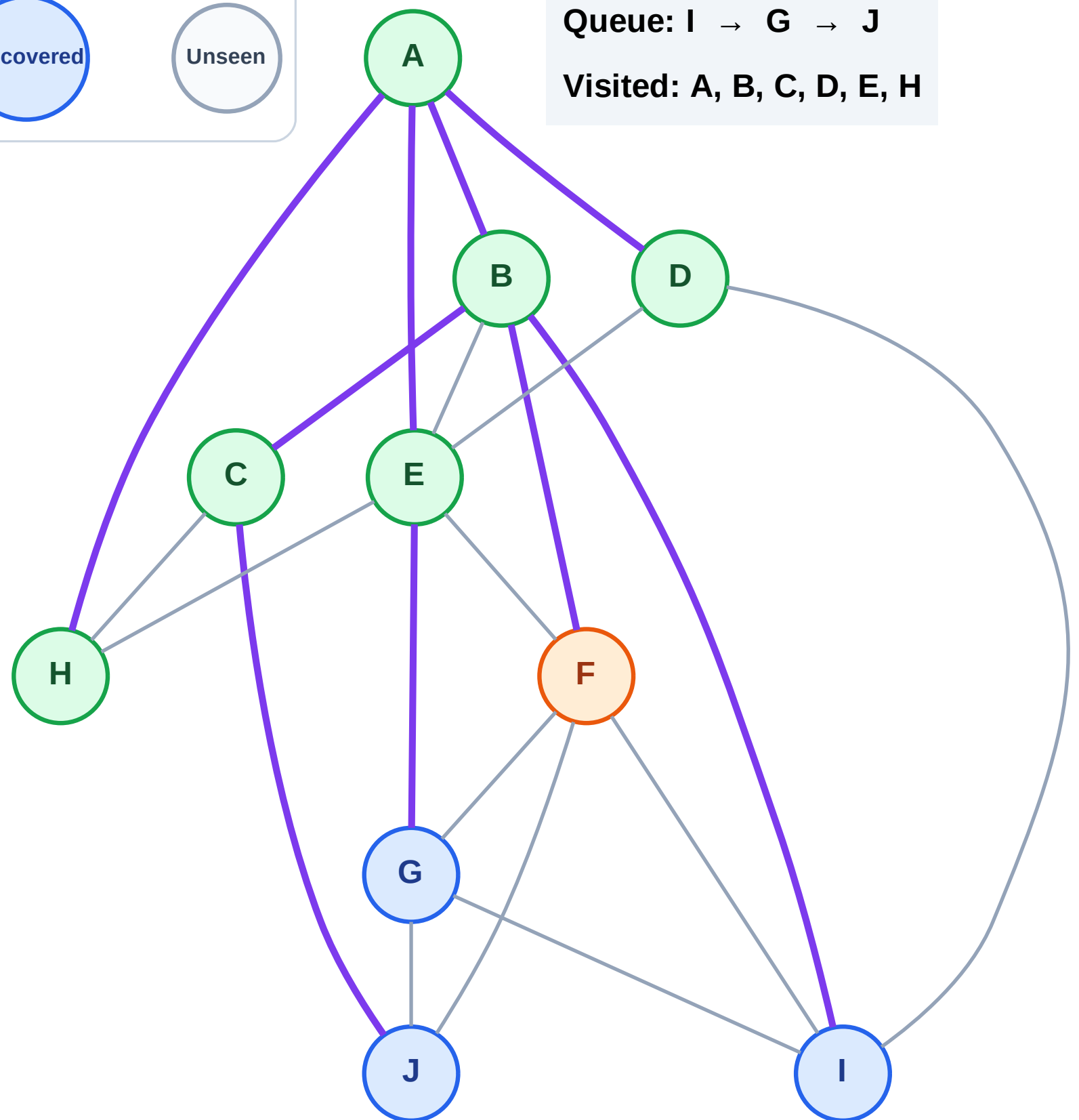
Step 14: Process node F

Legend



Queue: I → G → J

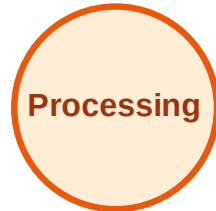
Visited: A, B, C, D, E, H



BFS Traversal

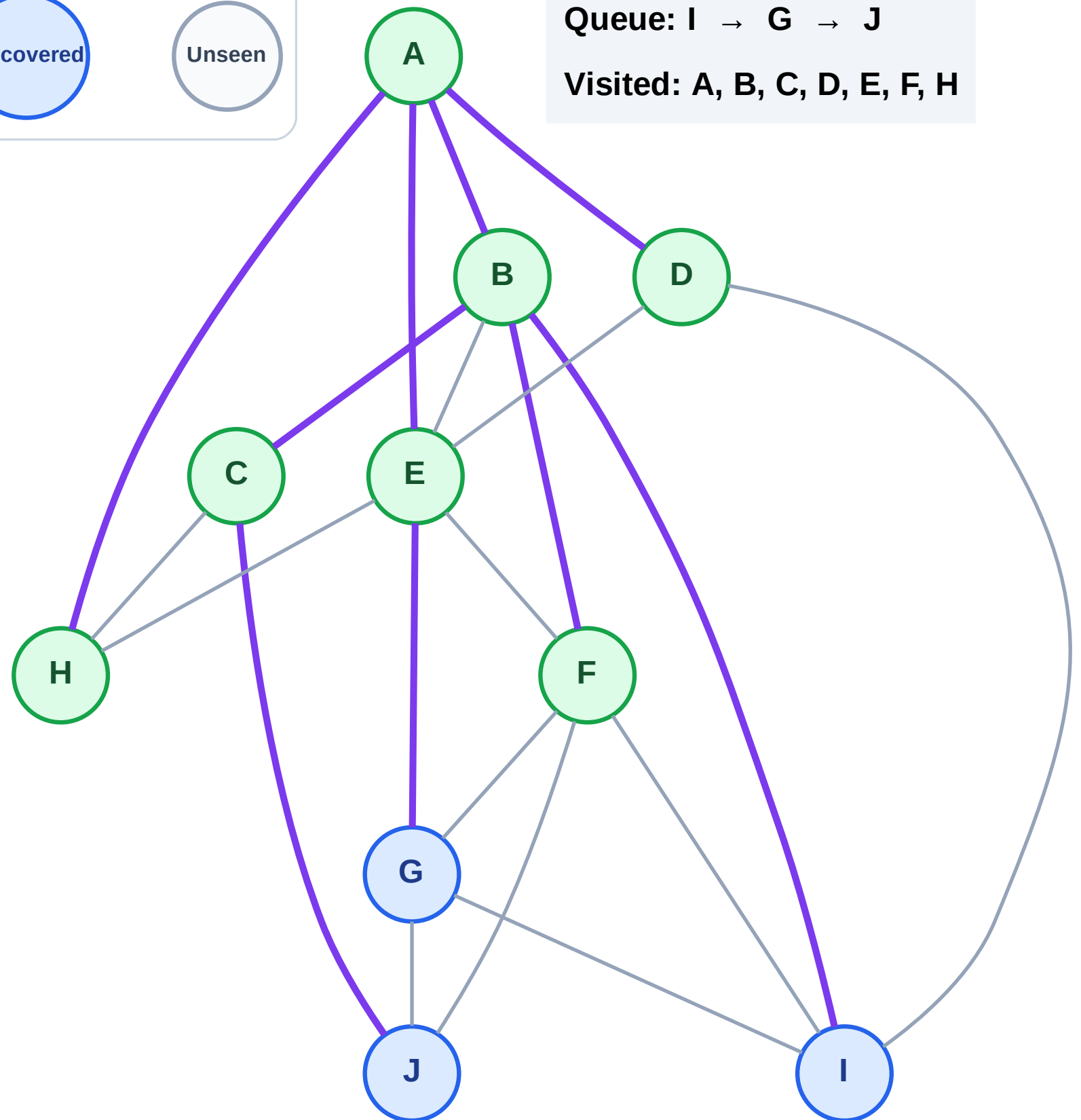
Step 15: No new neighbors from F

Legend



Queue: I → G → J

Visited: A, B, C, D, E, F, H



BFS Traversal

Step 16: Process node I

Legend

Complete

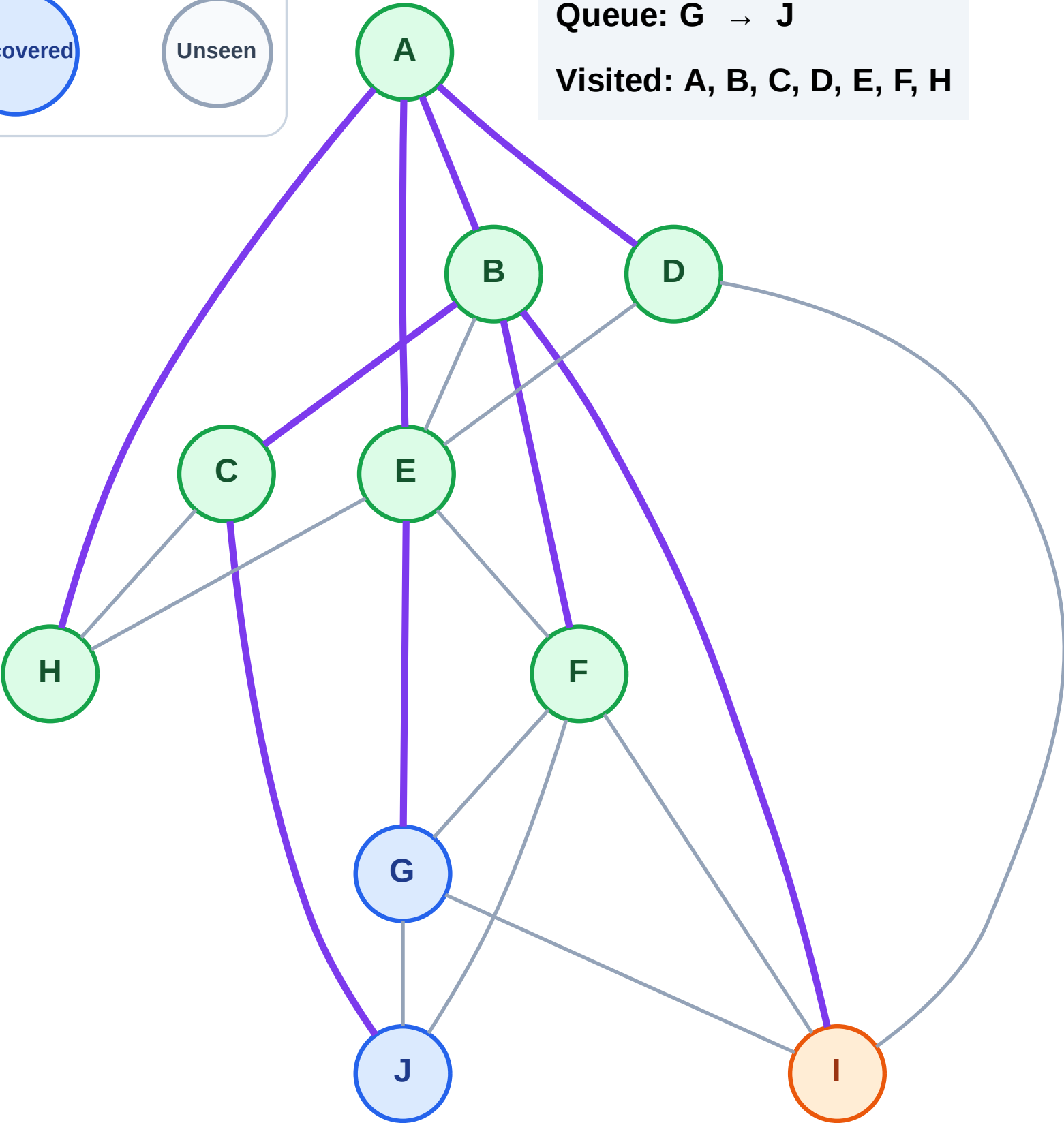
Processing

Discovered

Unseen

Queue: G → J

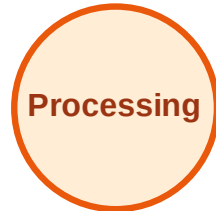
Visited: A, B, C, D, E, F, H



BFS Traversal

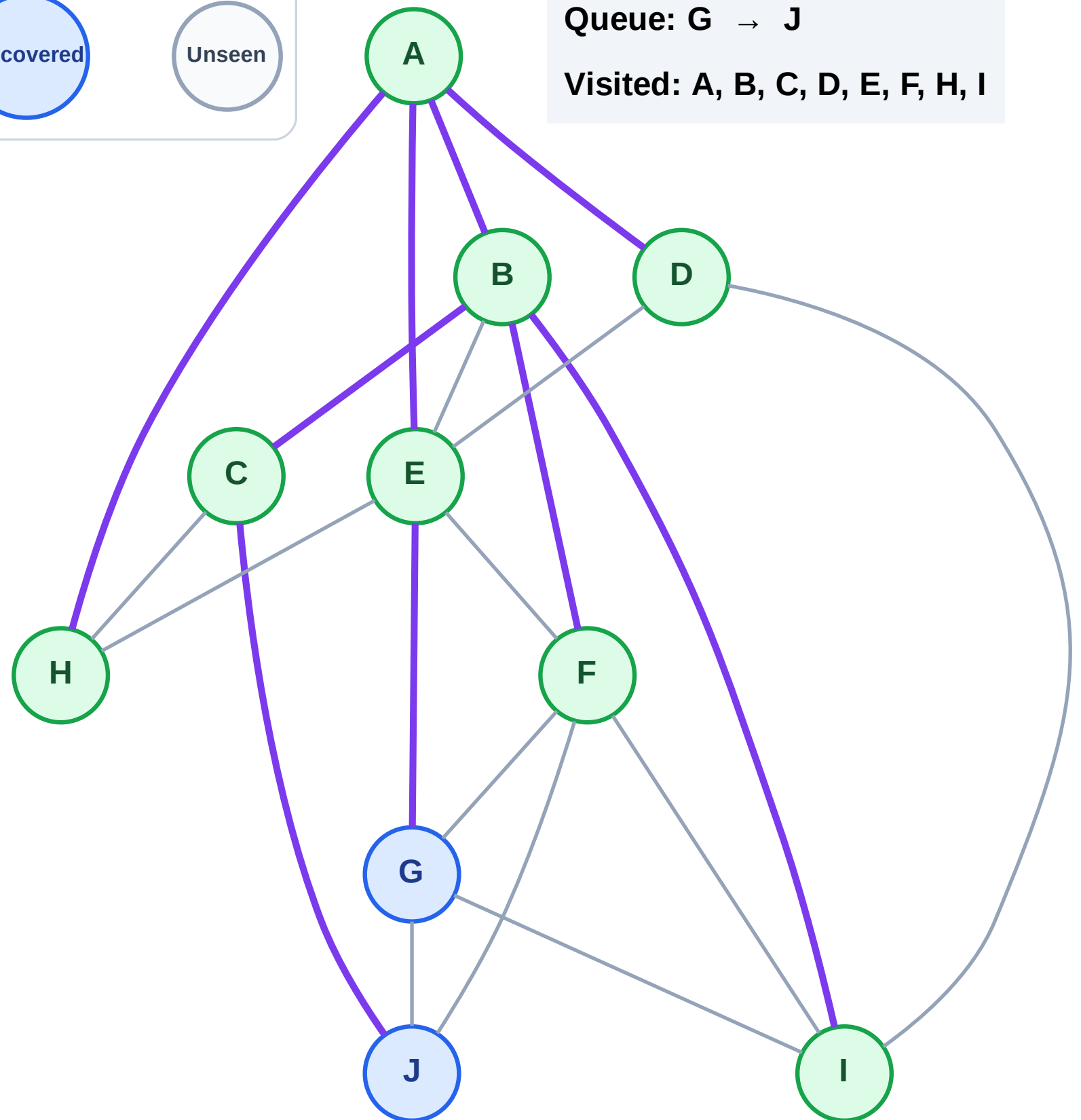
Step 17: No new neighbors from I

Legend



Queue: G → J

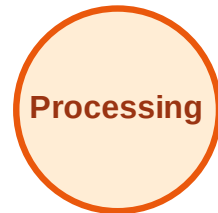
Visited: A, B, C, D, E, F, H, I



BFS Traversal

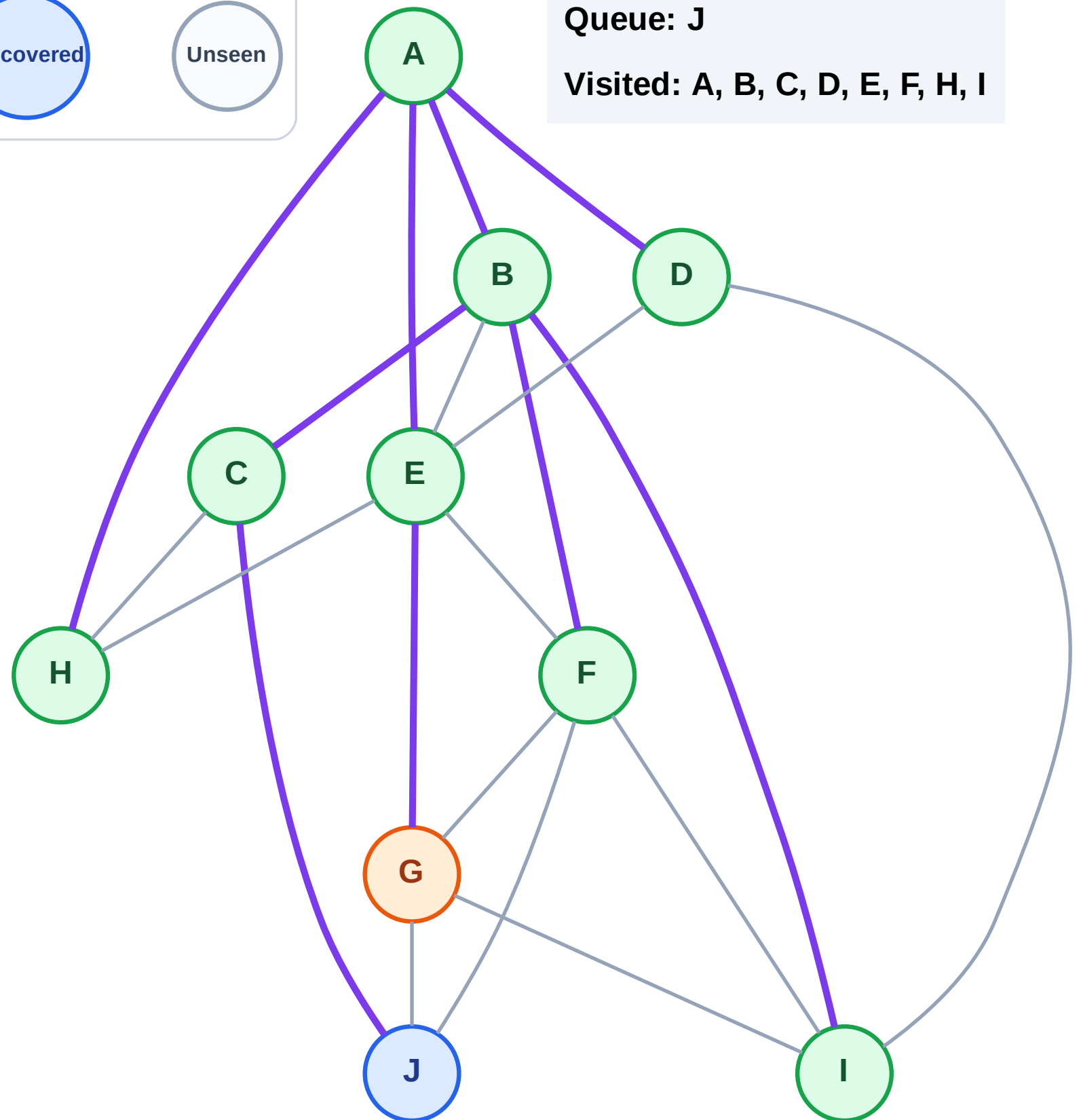
Step 18: Process node G

Legend



Queue: J

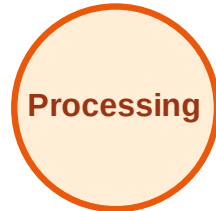
Visited: A, B, C, D, E, F, H, I



BFS Traversal

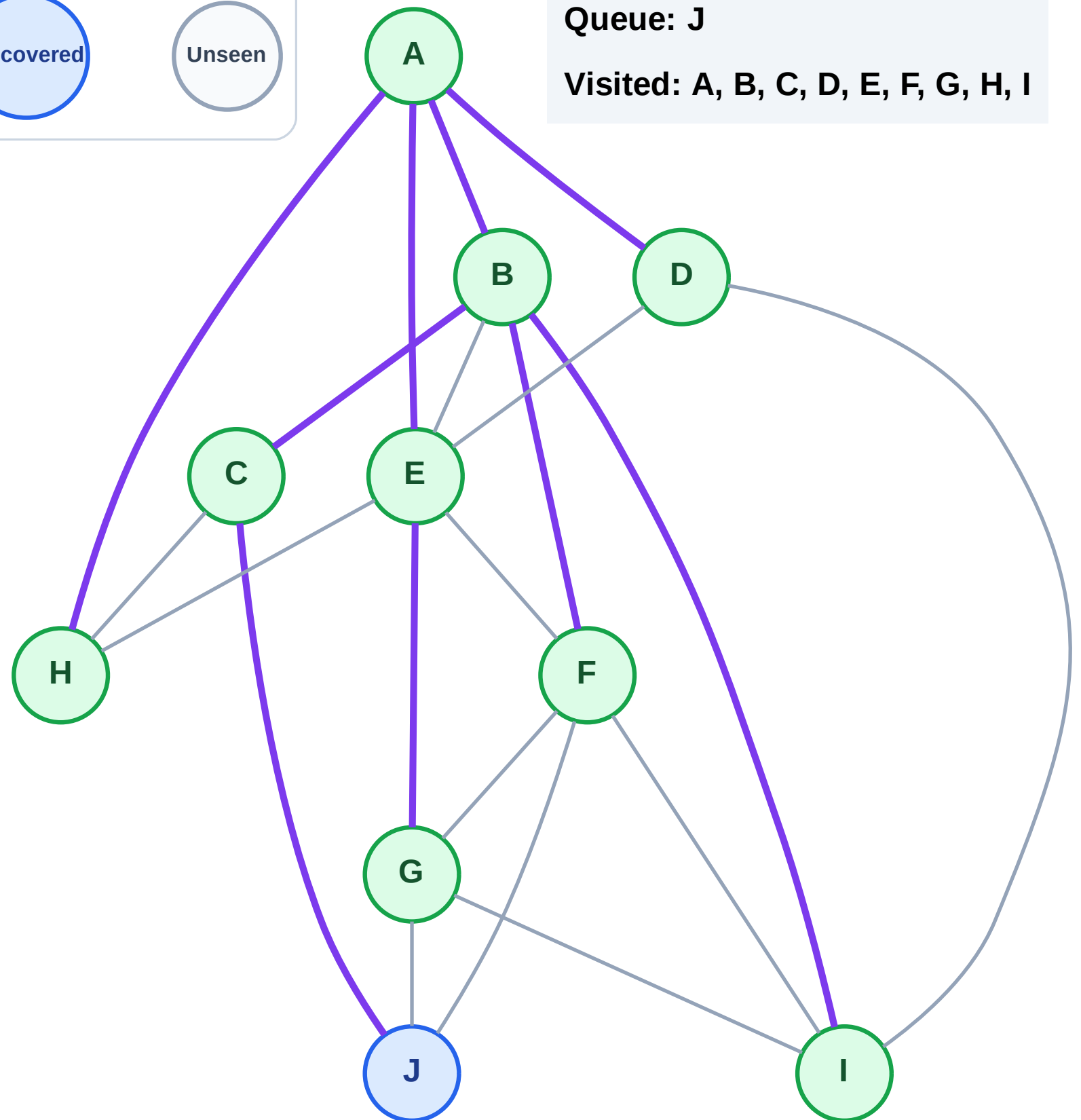
Step 19: No new neighbors from G

Legend



Queue: J

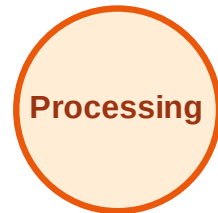
Visited: A, B, C, D, E, F, G, H, I



BFS Traversal

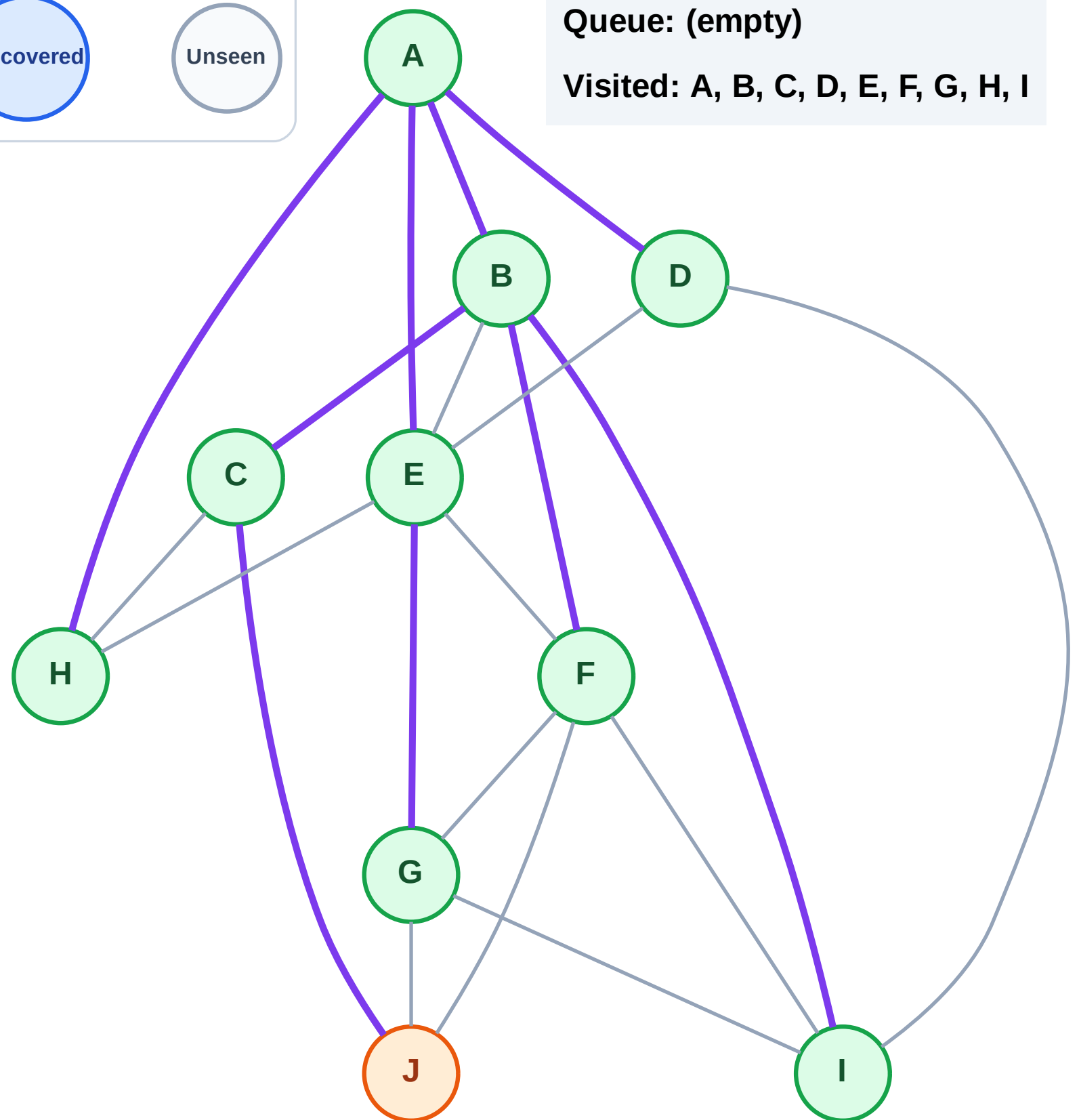
Step 20: Process node J

Legend



Queue: (empty)

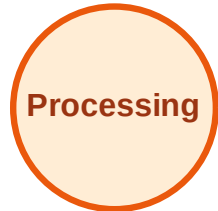
Visited: A, B, C, D, E, F, G, H, I



BFS Traversal

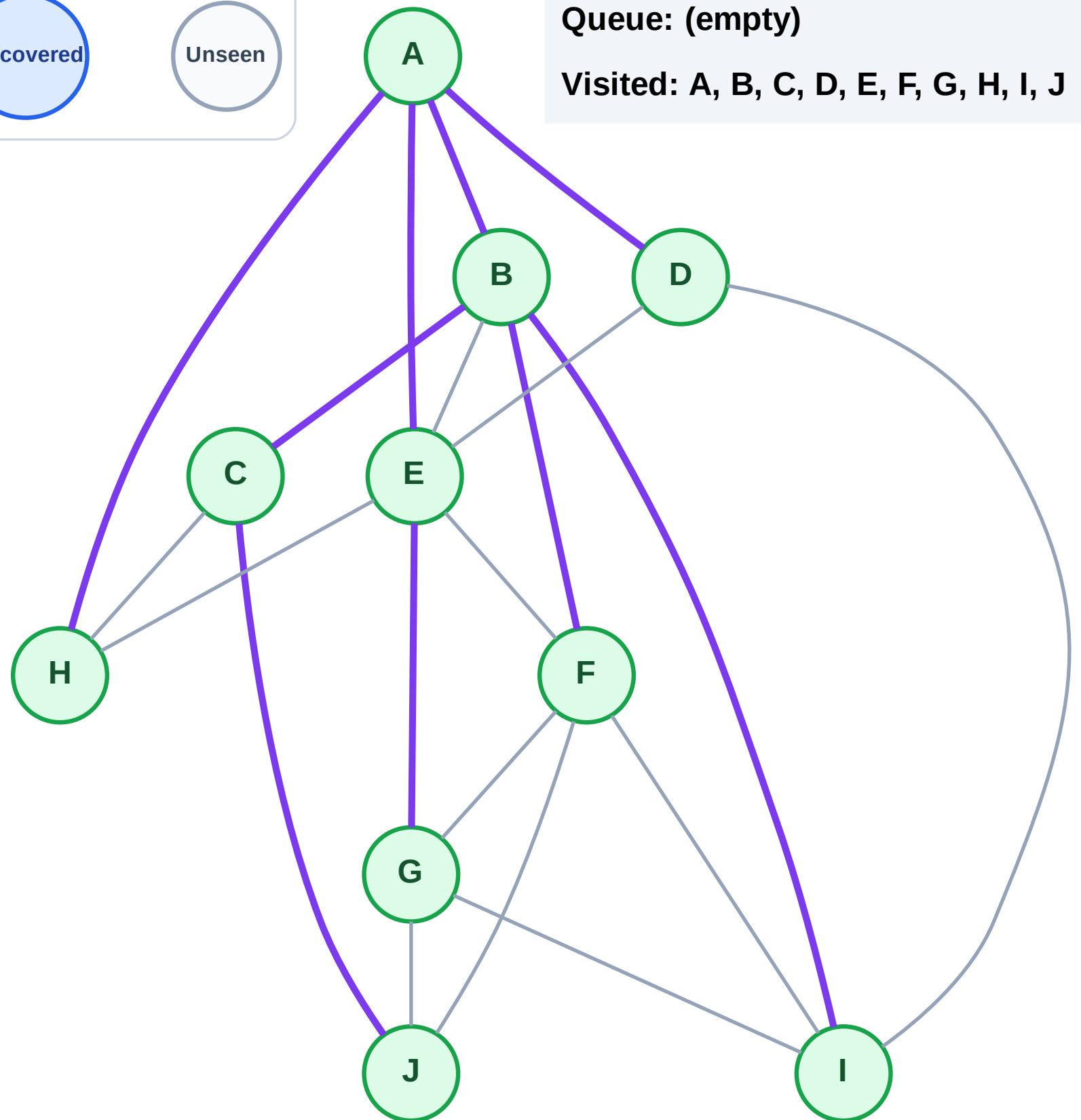
Step 21: No new neighbors from J

Legend



Queue: (empty)

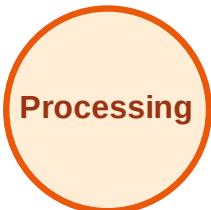
Visited: A, B, C, D, E, F, G, H, I, J



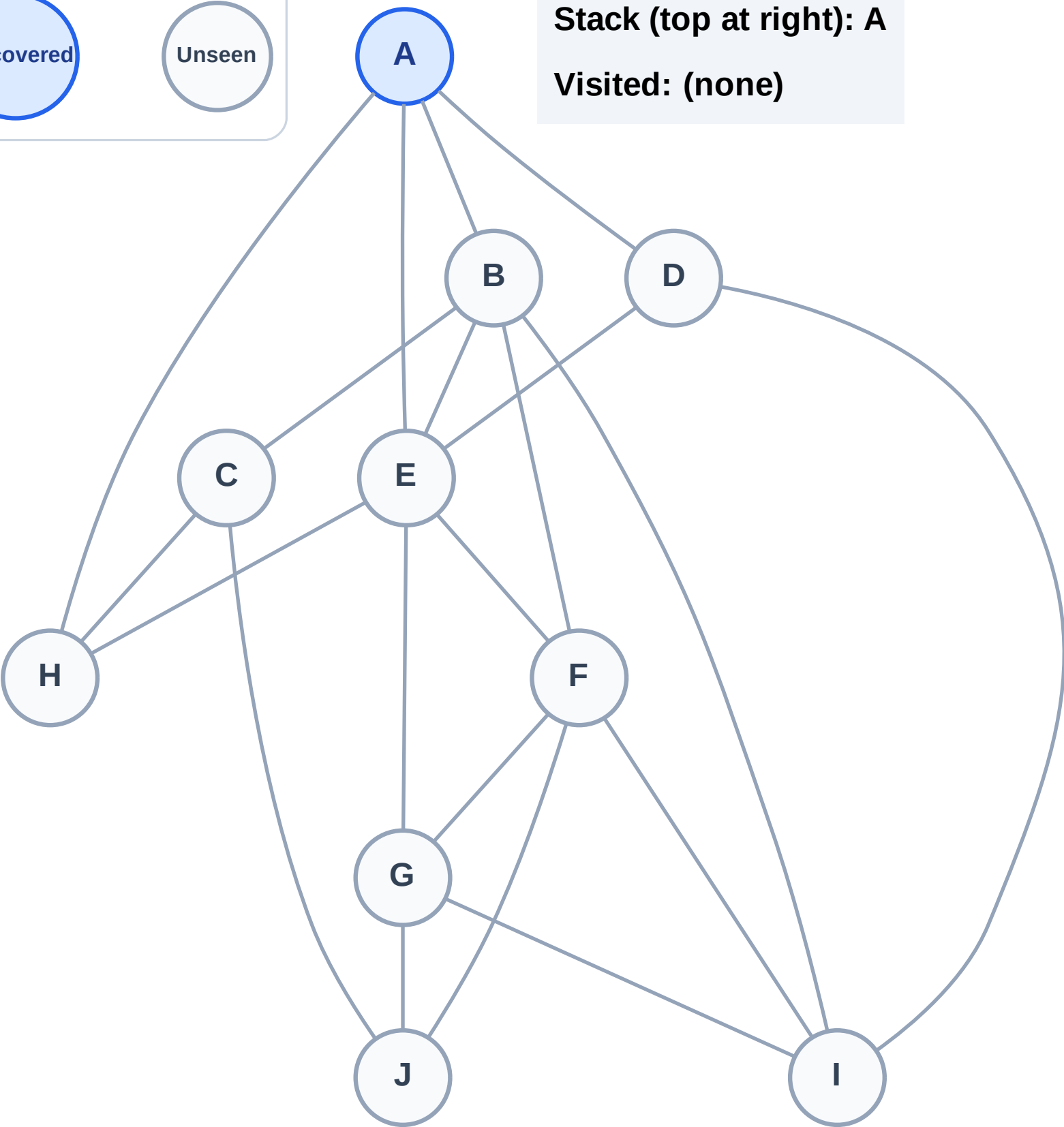
DFS Traversal

Step 1: Start at A

Legend



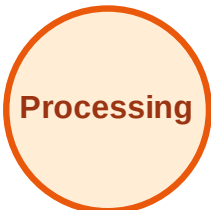
Stack (top at right): A
Visited: (none)



DFS Traversal

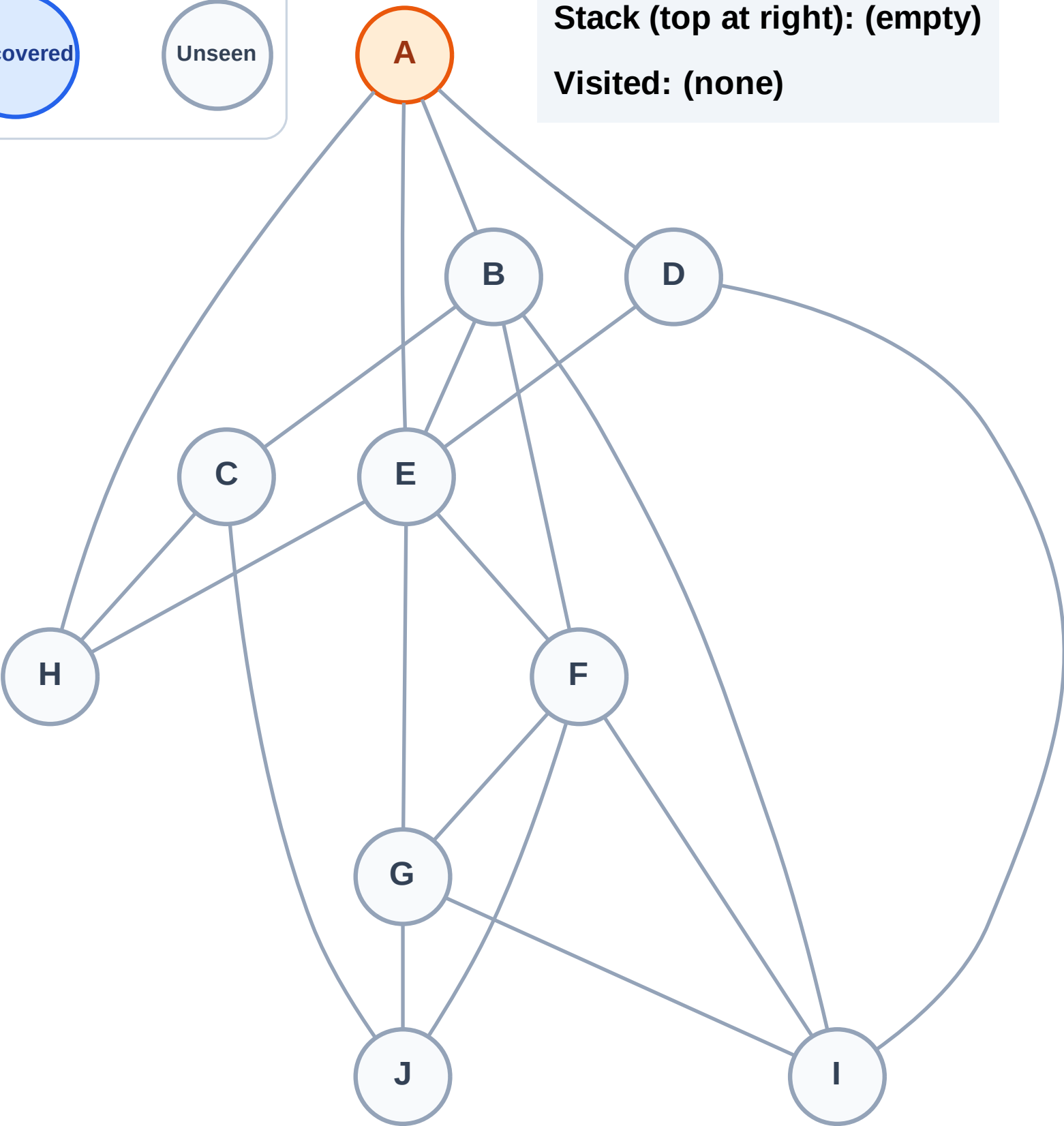
Step 2: Process node A

Legend



Stack (top at right): (empty)

Visited: (none)



DFS Traversal

Step 3: Discover H, E, D, B from A

Legend

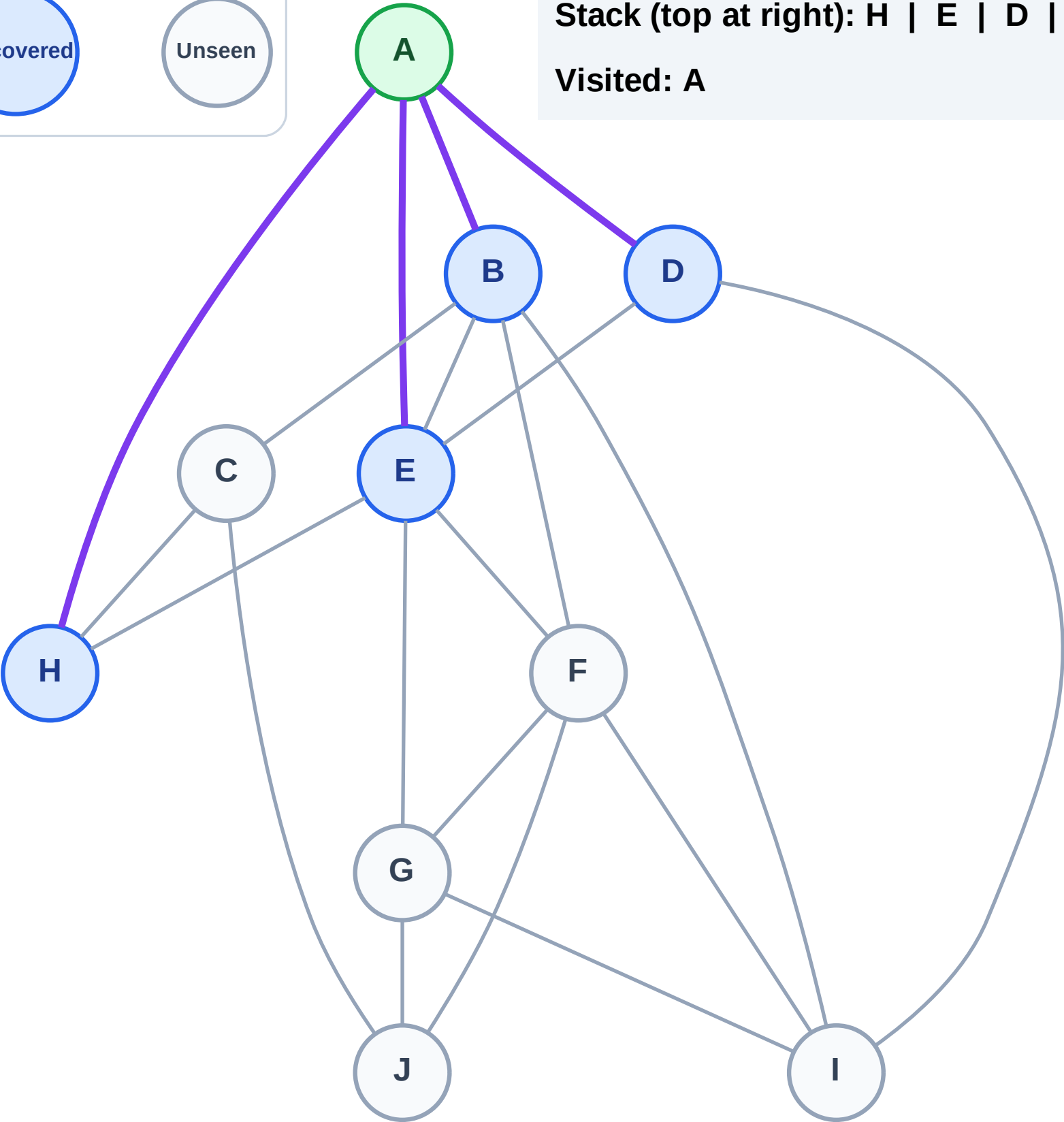
Complete

Processing

Discovered

Unseen

Stack (top at right): H | E | D | B
Visited: A



DFS Traversal

Step 4: Process node B

Legend

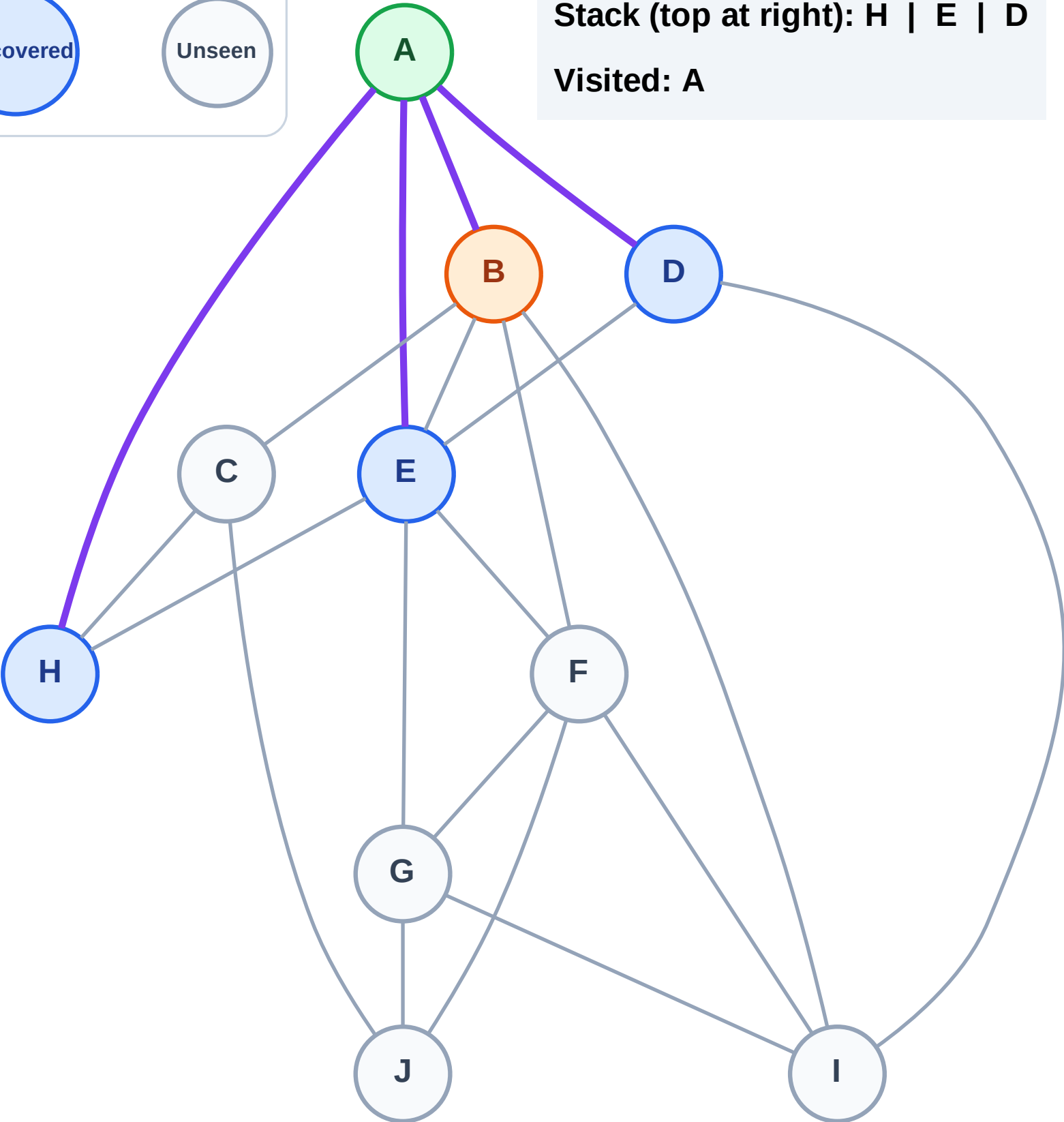
Complete

Processing

Discovered

Unseen

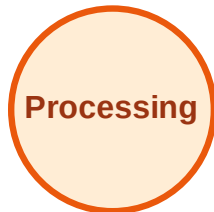
Stack (top at right): H | E | D
Visited: A



DFS Traversal

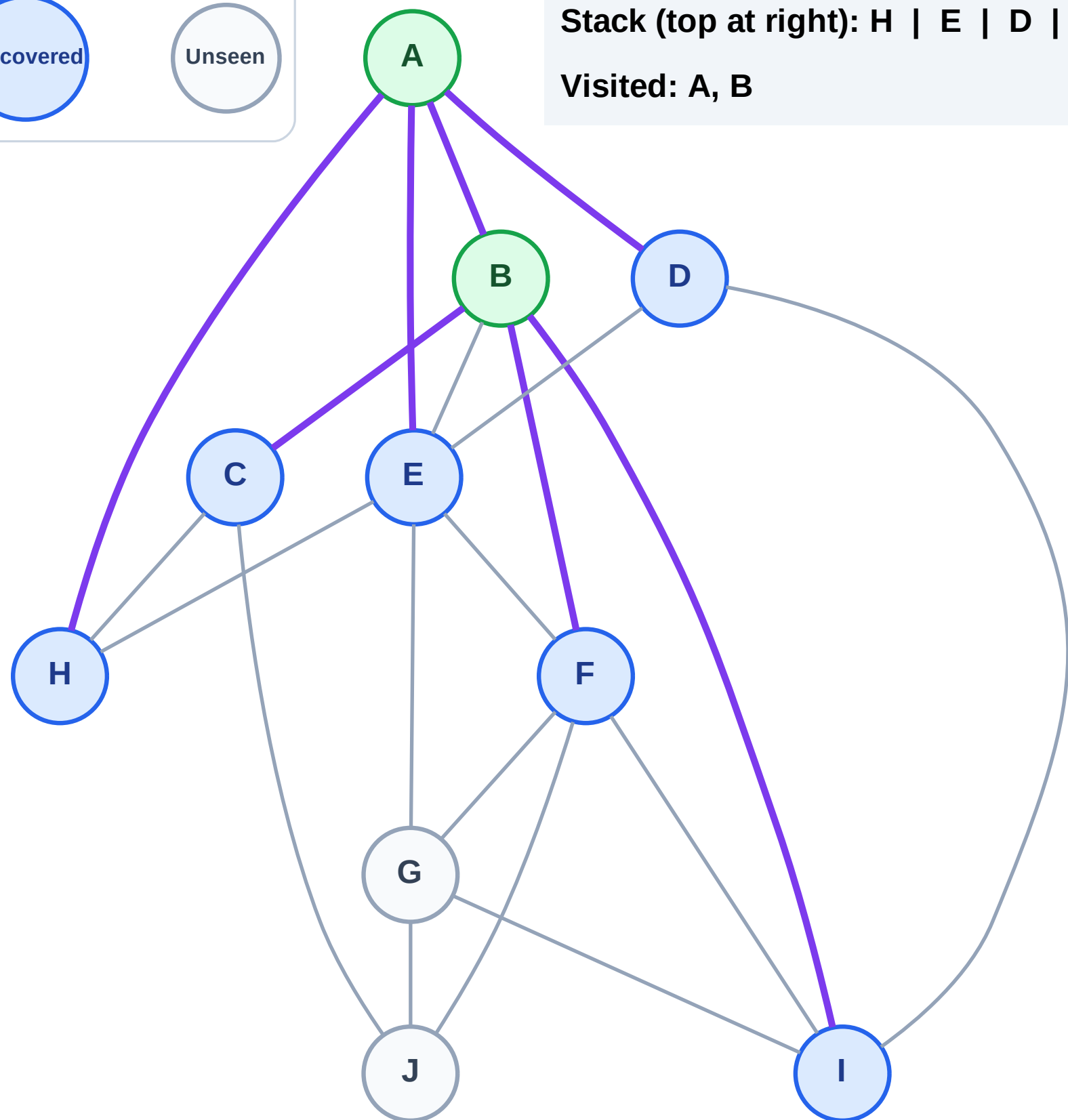
Step 5: Discover I, F, C from B

Legend



Stack (top at right): H | E | D | I | F | C

Visited: A, B



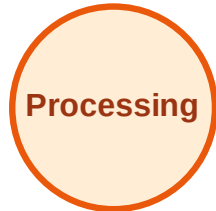
DFS Traversal

Step 6: Process node C

Legend



Complete



Processing



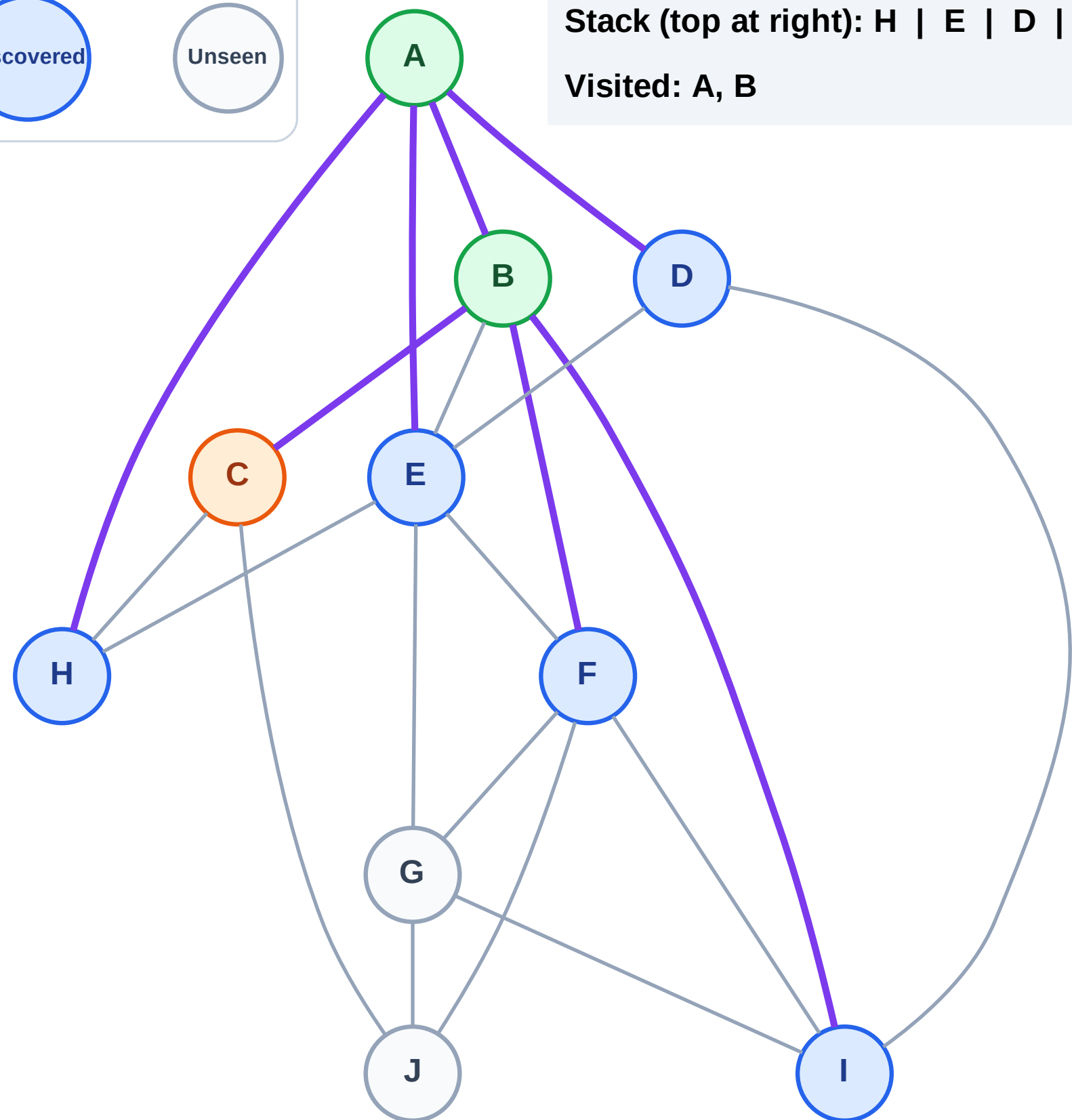
Discovered



Unseen

Stack (top at right): H | E | D | I | F

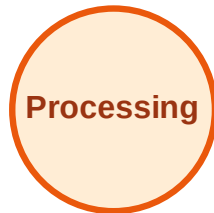
Visited: A, B



DFS Traversal

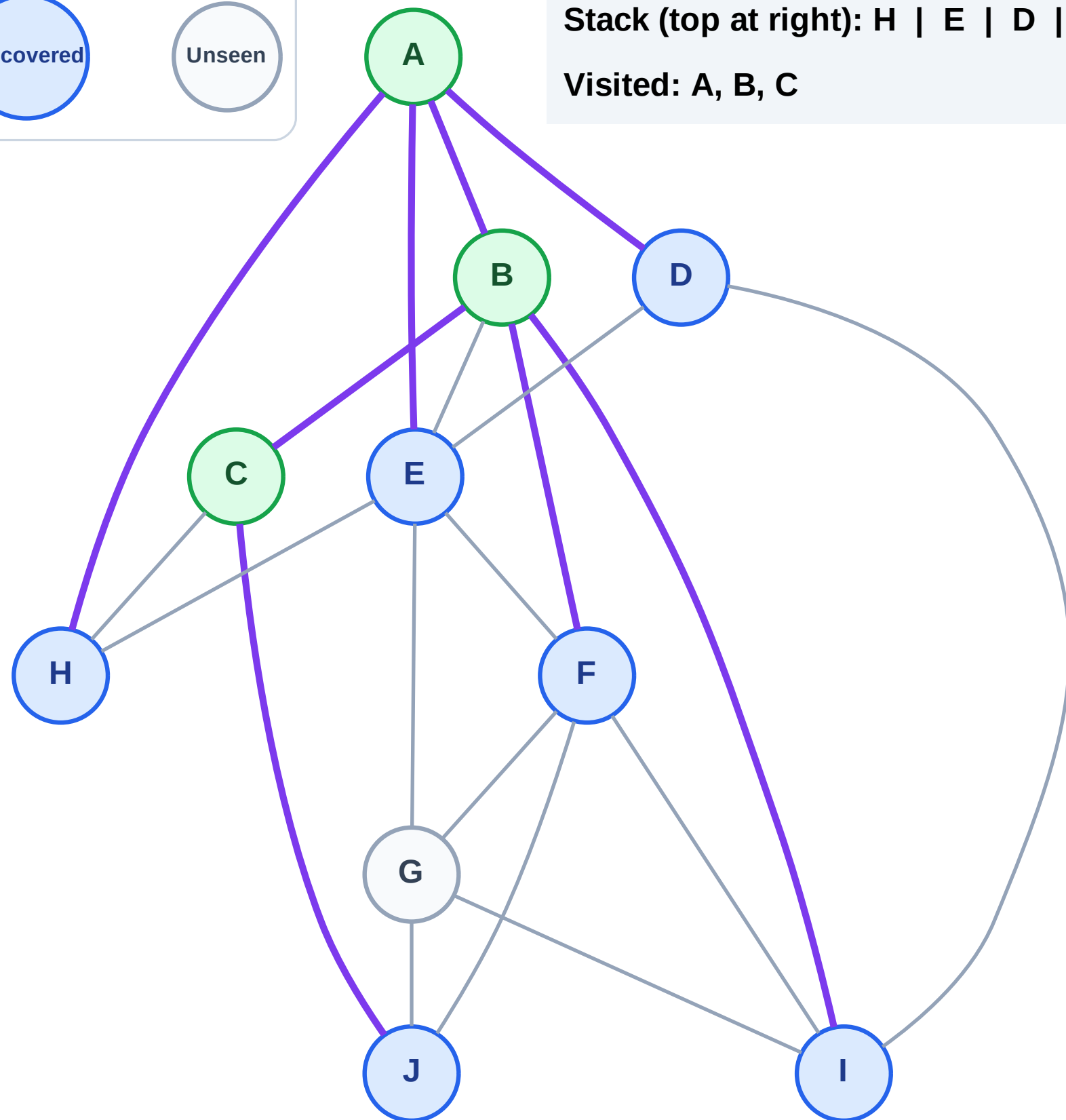
Step 7: Discover J from C

Legend



Stack (top at right): H | E | D | I | F | J

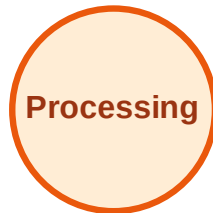
Visited: A, B, C



DFS Traversal

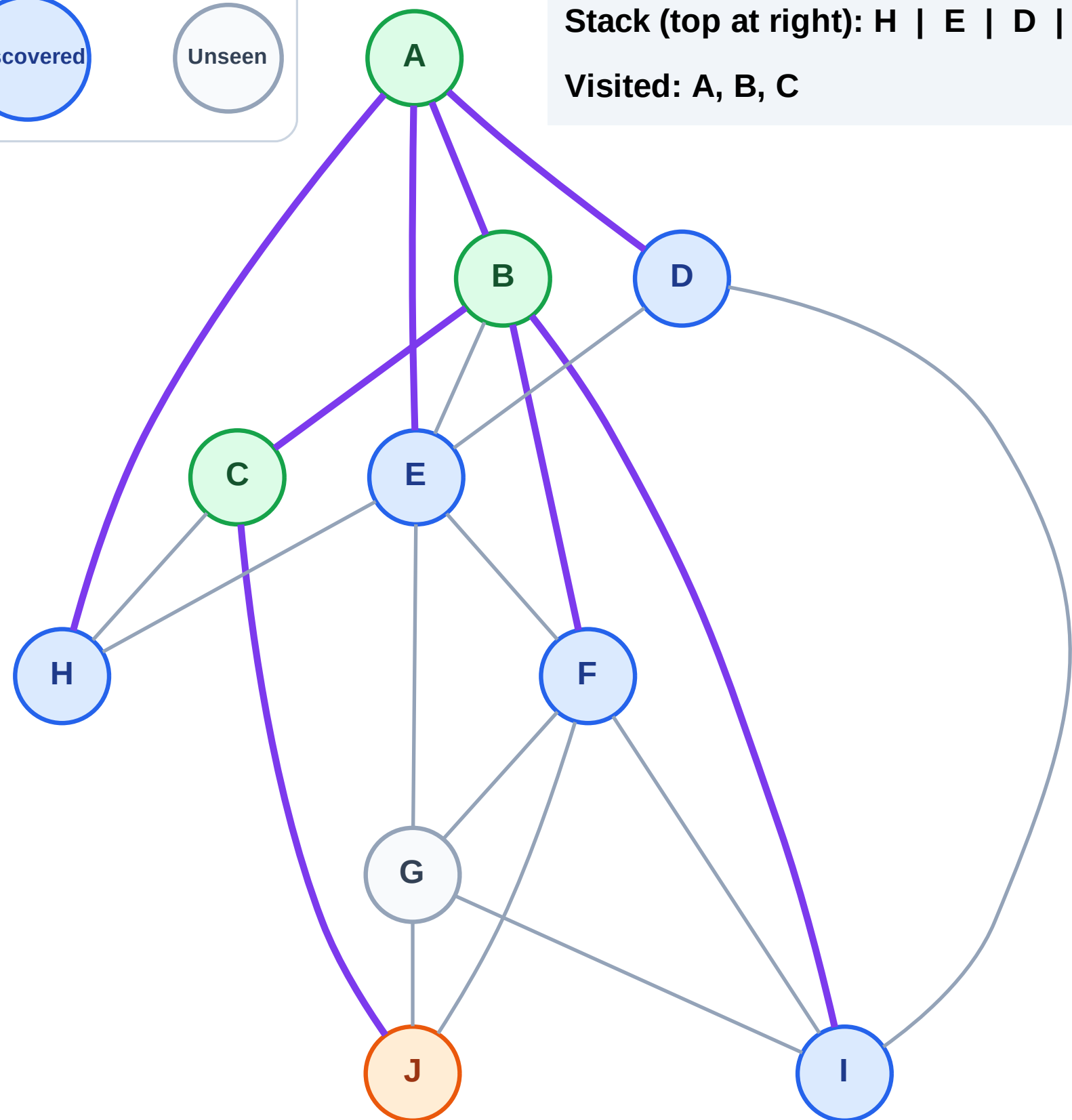
Step 8: Process node J

Legend



Stack (top at right): H | E | D | I | F

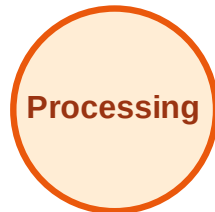
Visited: A, B, C



DFS Traversal

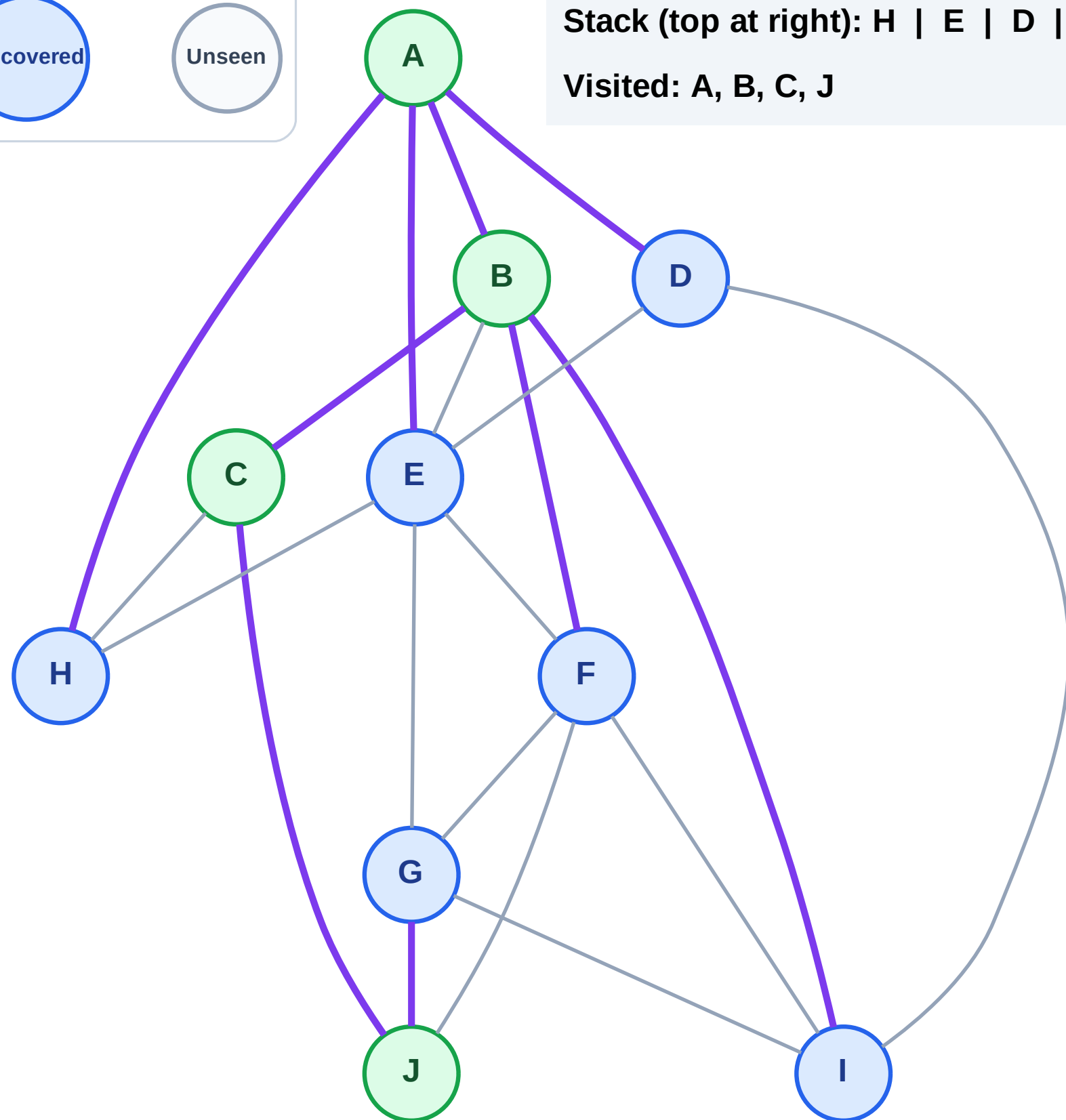
Step 9: Discover G from J

Legend



Stack (top at right): H | E | D | I | F | G

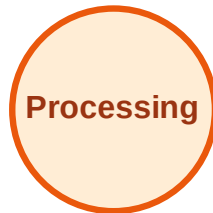
Visited: A, B, C, J



DFS Traversal

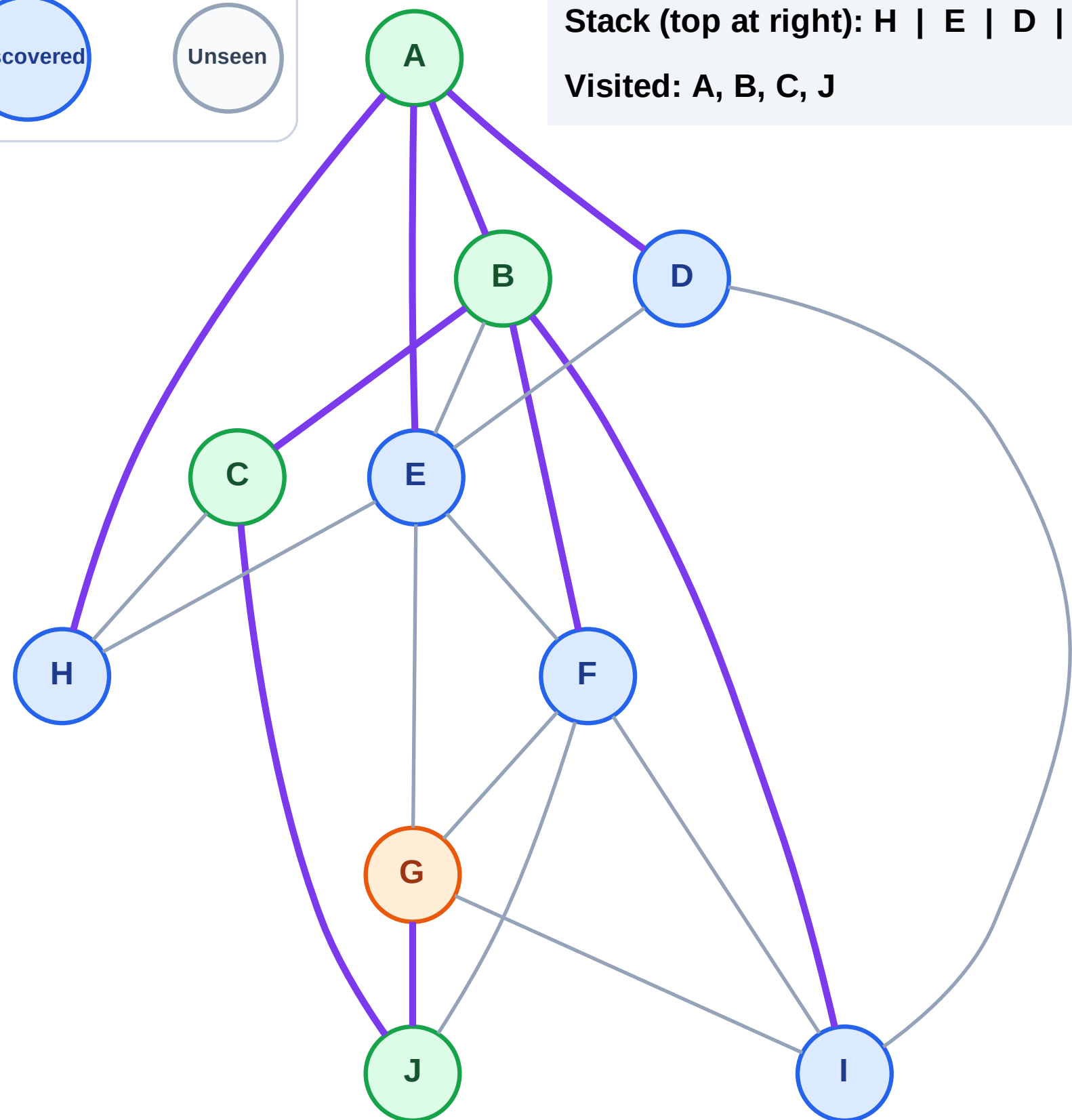
Step 10: Process node G

Legend



Stack (top at right): H | E | D | I | F

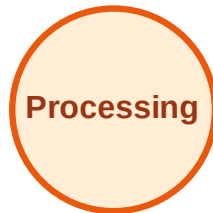
Visited: A, B, C, J



DFS Traversal

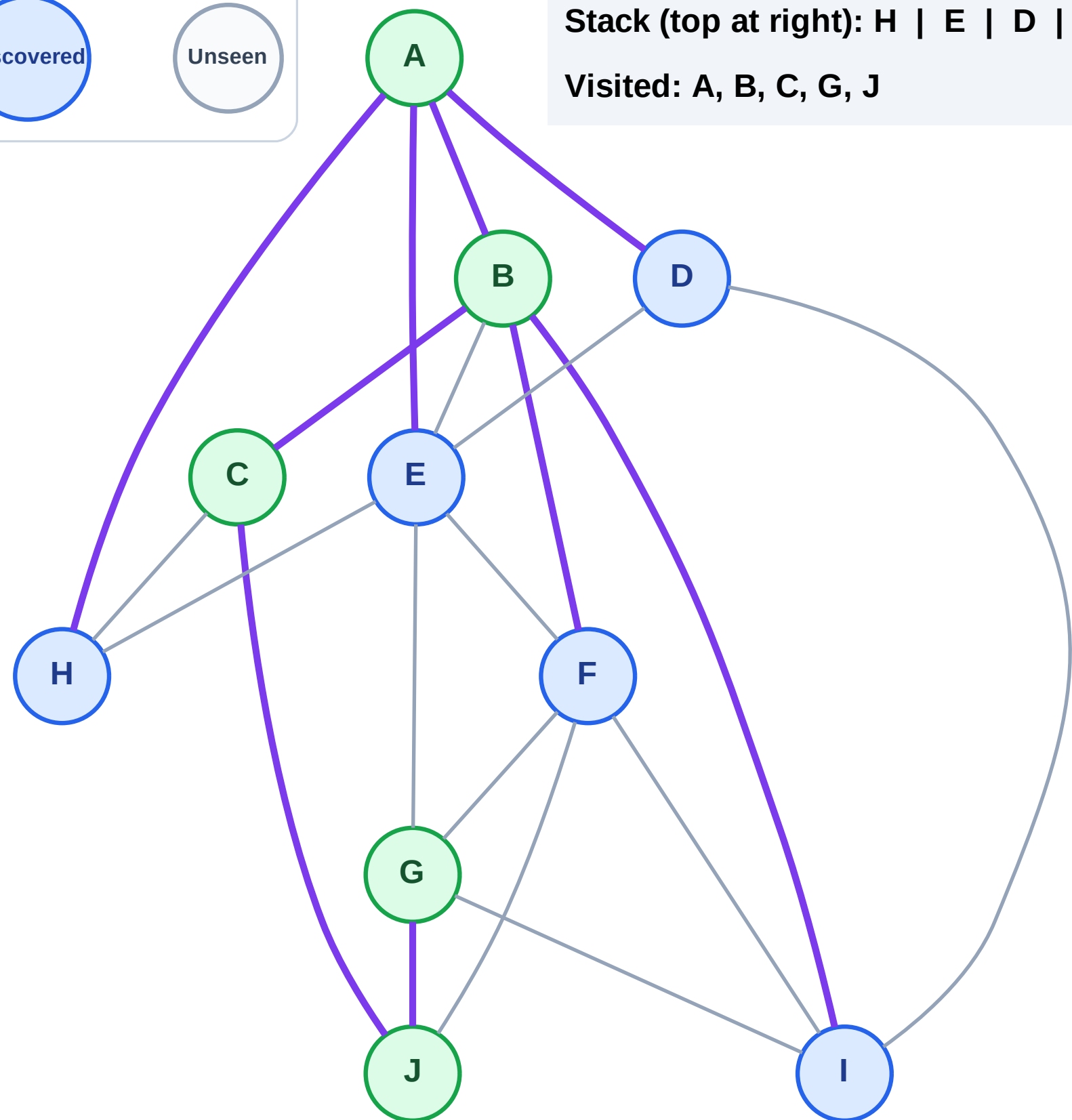
Step 11: No new neighbors from G

Legend



Stack (top at right): H | E | D | I | F

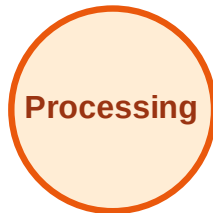
Visited: A, B, C, G, J



DFS Traversal

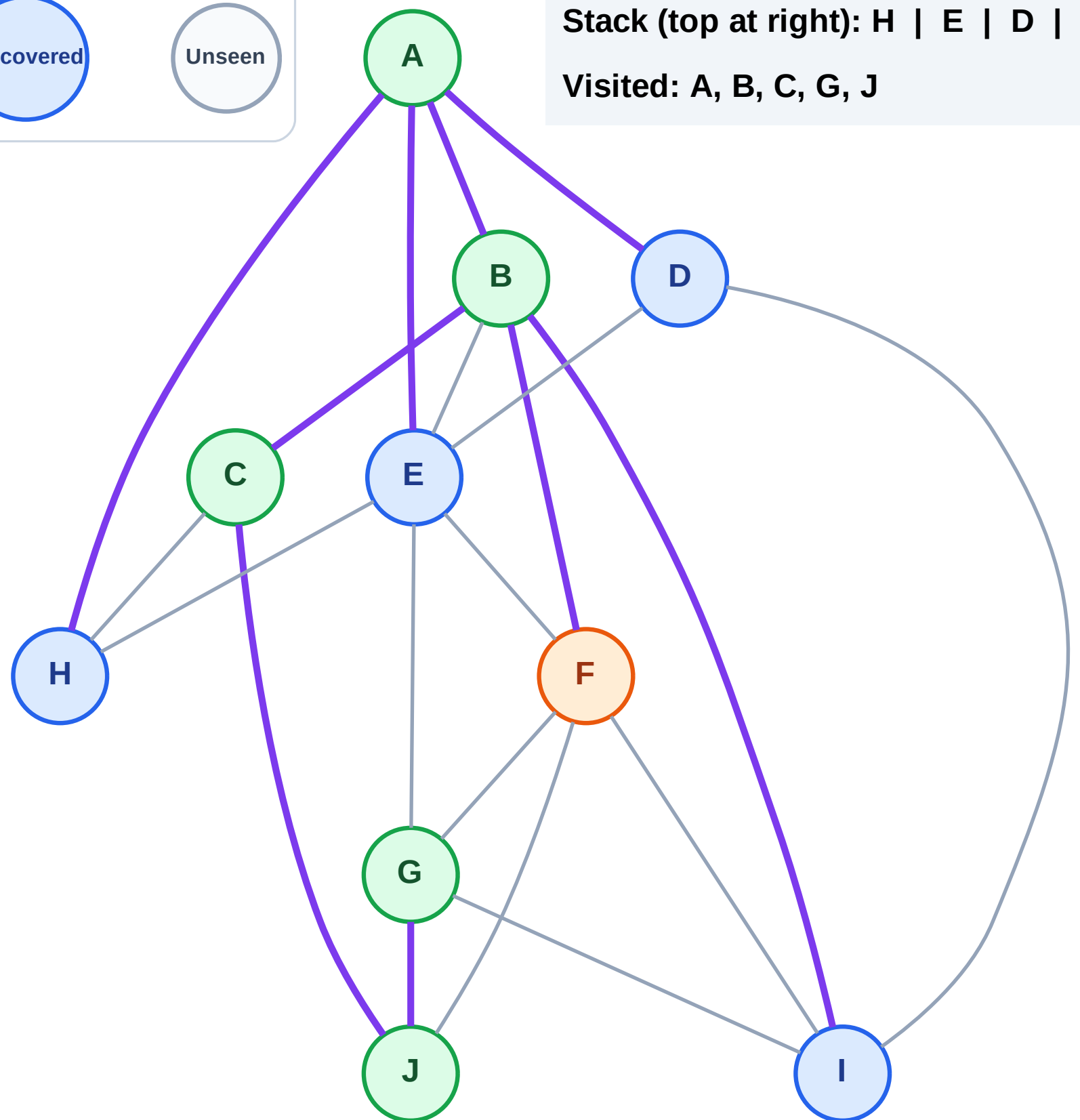
Step 12: Process node F

Legend



Stack (top at right): H | E | D | I

Visited: A, B, C, G, J



DFS Traversal

Step 13: No new neighbors from F

Legend

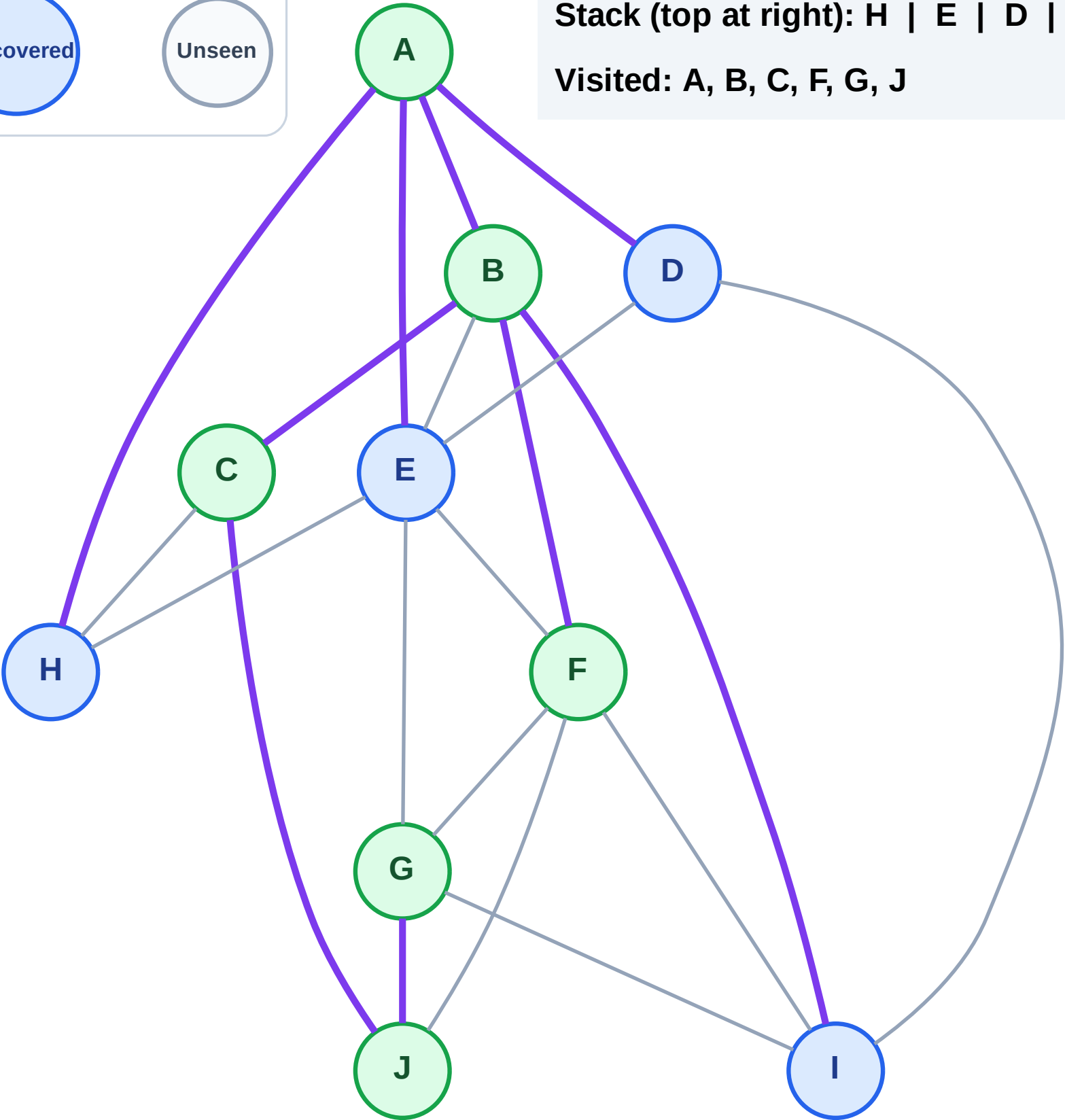
Complete

Processing

Discovered

Unseen

Stack (top at right): H | E | D | I
Visited: A, B, C, F, G, J



DFS Traversal

Step 14: Process node I

Legend

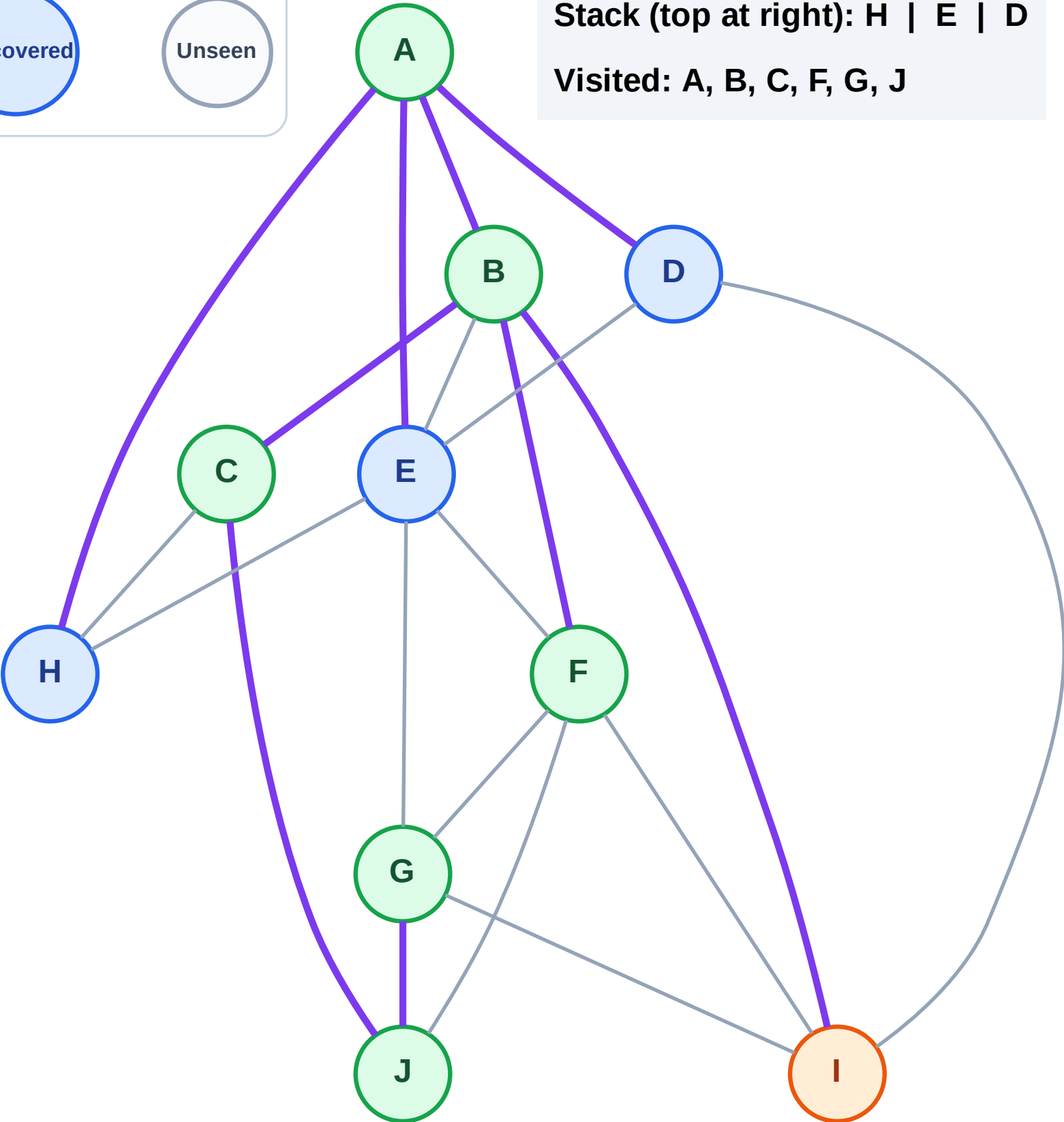
Complete

Processing

Discovered

Unseen

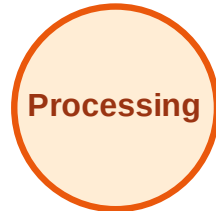
Stack (top at right): H | E | D
Visited: A, B, C, F, G, J



DFS Traversal

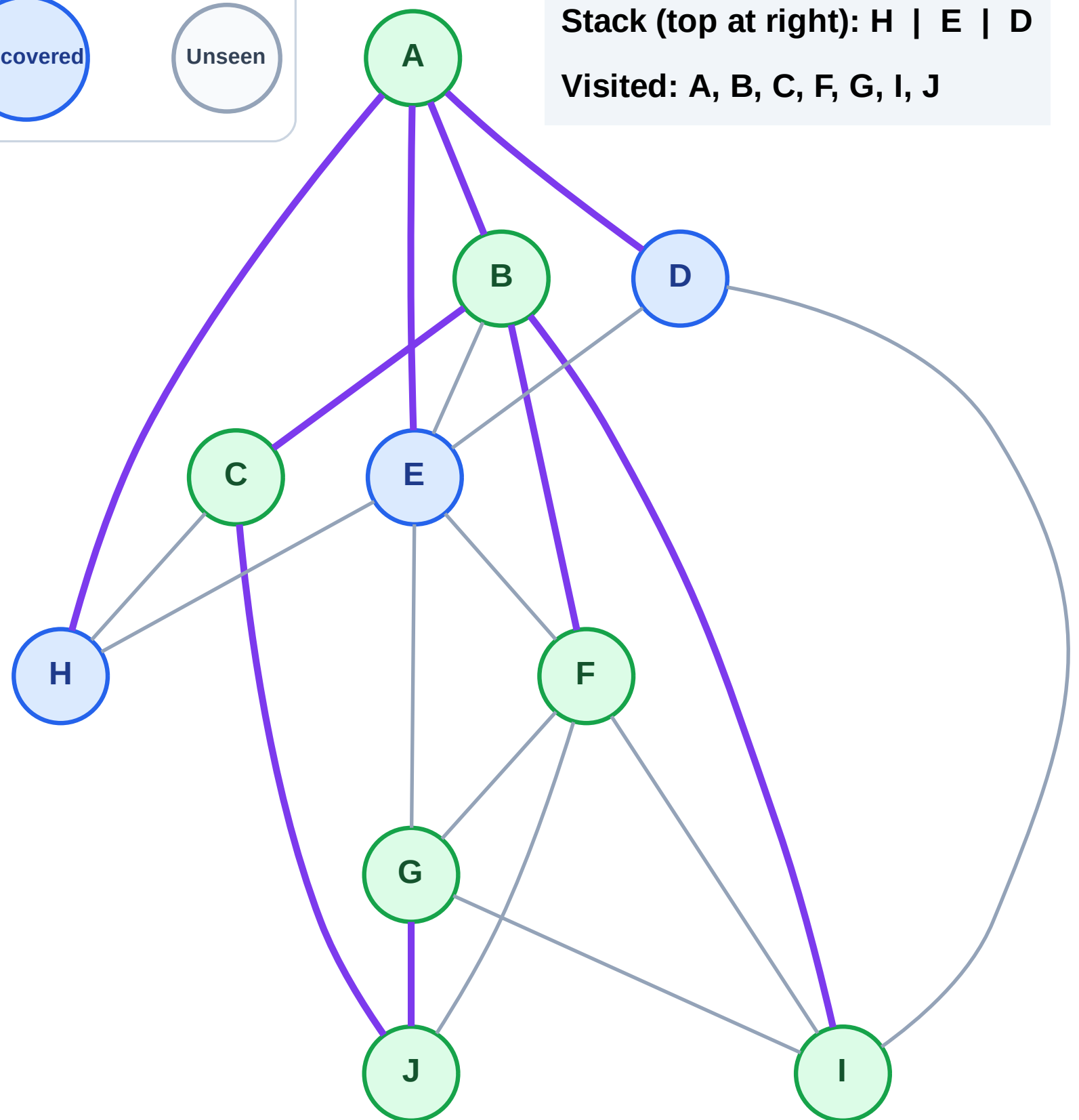
Step 15: No new neighbors from I

Legend



Stack (top at right): H | E | D

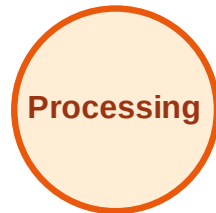
Visited: A, B, C, F, G, I, J



DFS Traversal

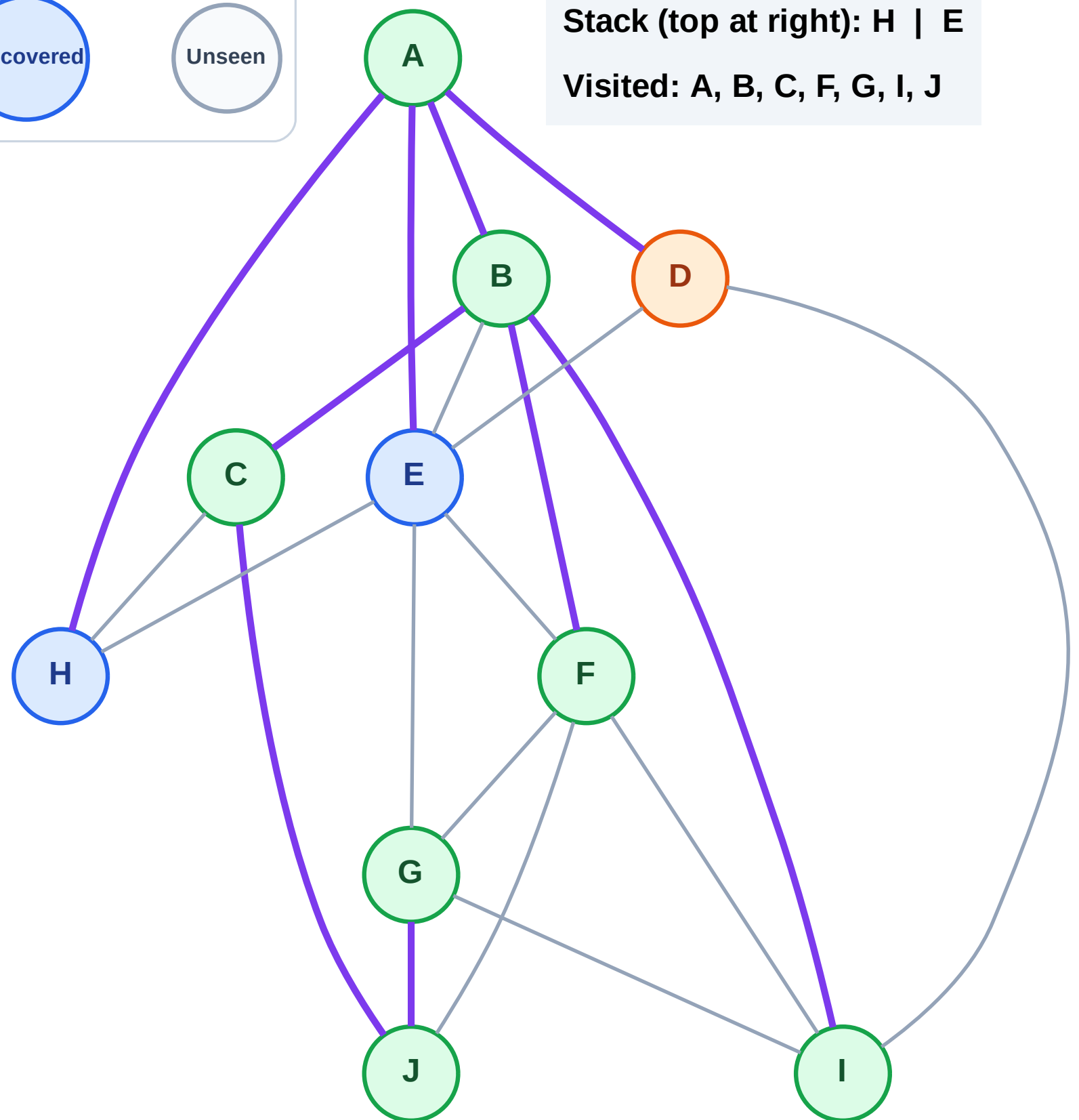
Step 16: Process node D

Legend



Stack (top at right): H | E

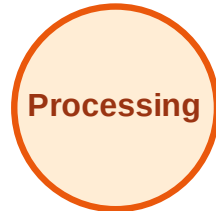
Visited: A, B, C, F, G, I, J



DFS Traversal

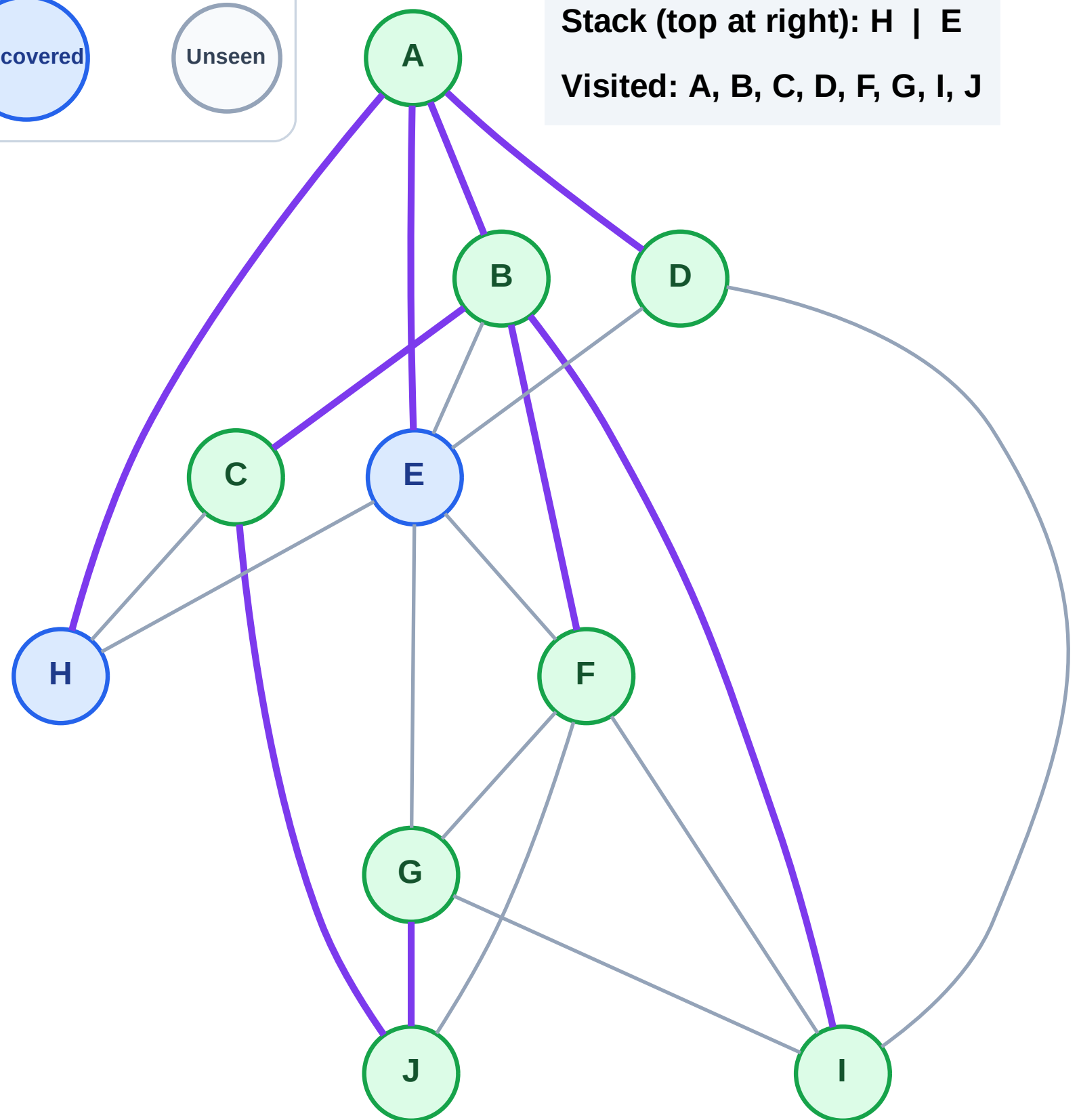
Step 17: No new neighbors from D

Legend



Stack (top at right): H | E

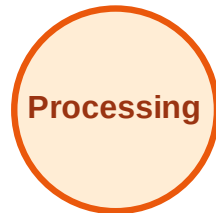
Visited: A, B, C, D, F, G, I, J



DFS Traversal

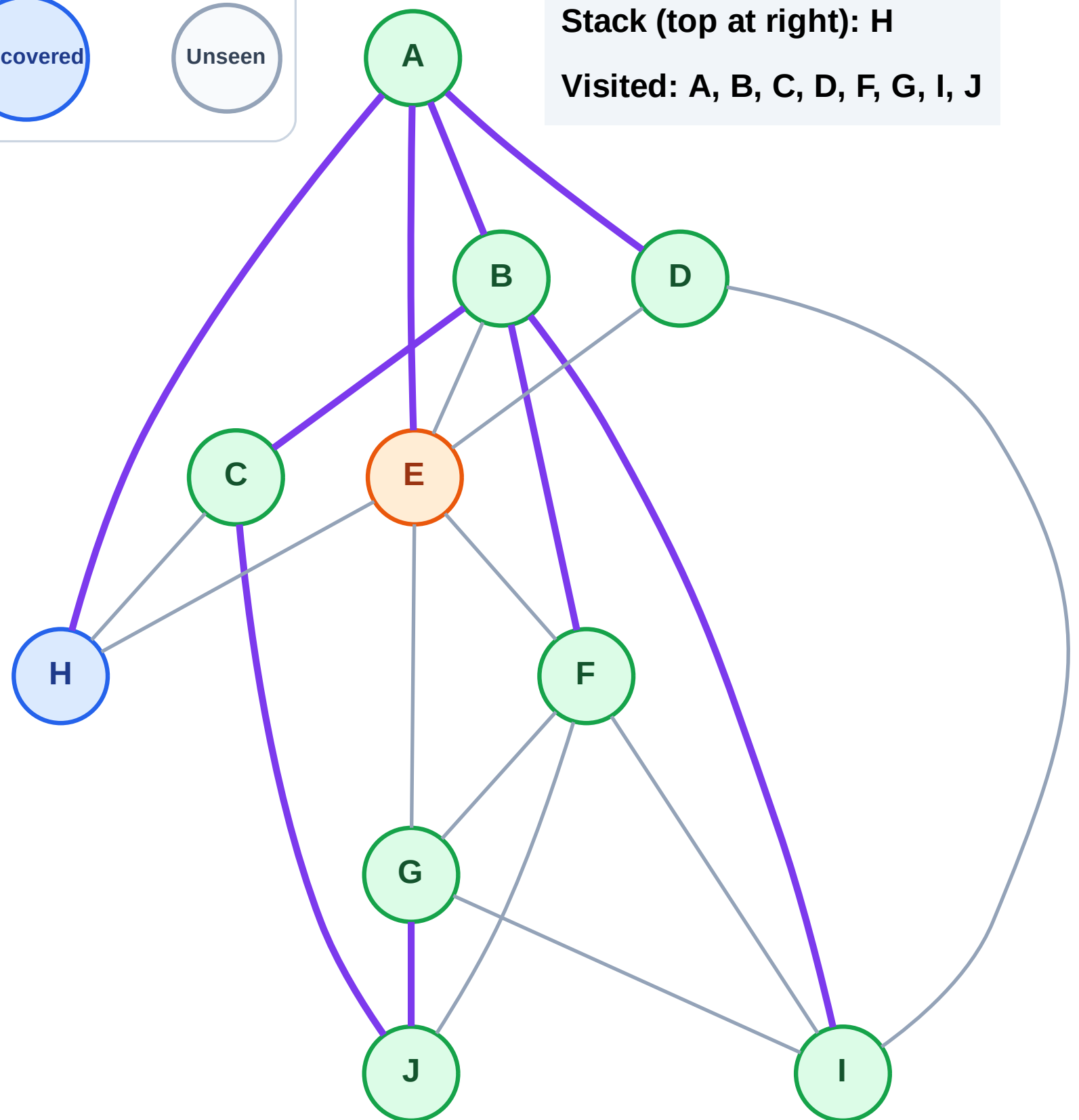
Step 18: Process node E

Legend



Stack (top at right): H

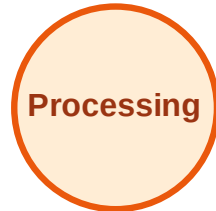
Visited: A, B, C, D, F, G, I, J



DFS Traversal

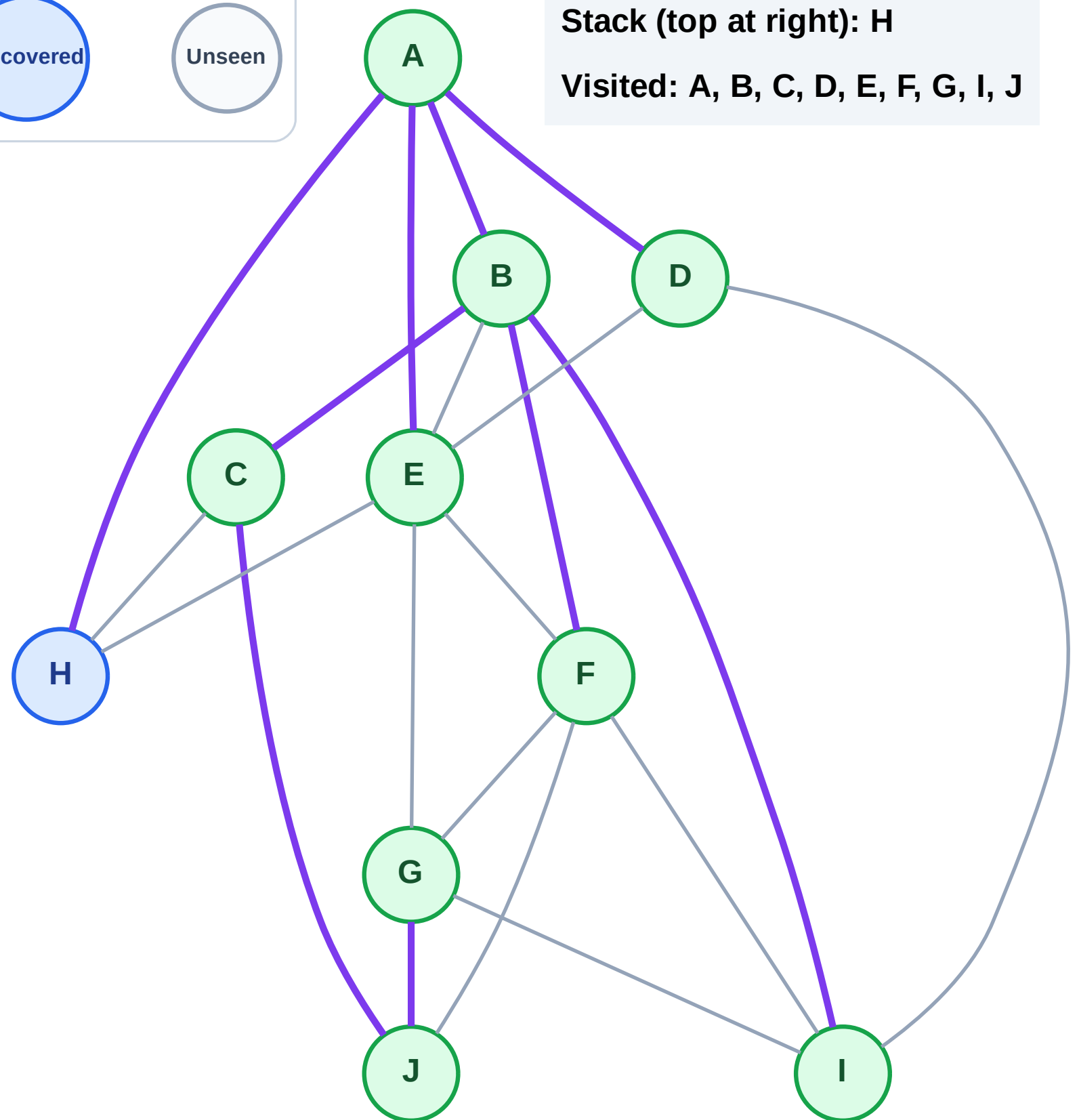
Step 19: No new neighbors from E

Legend



Stack (top at right): H

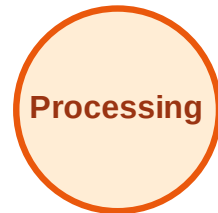
Visited: A, B, C, D, E, F, G, I, J



DFS Traversal

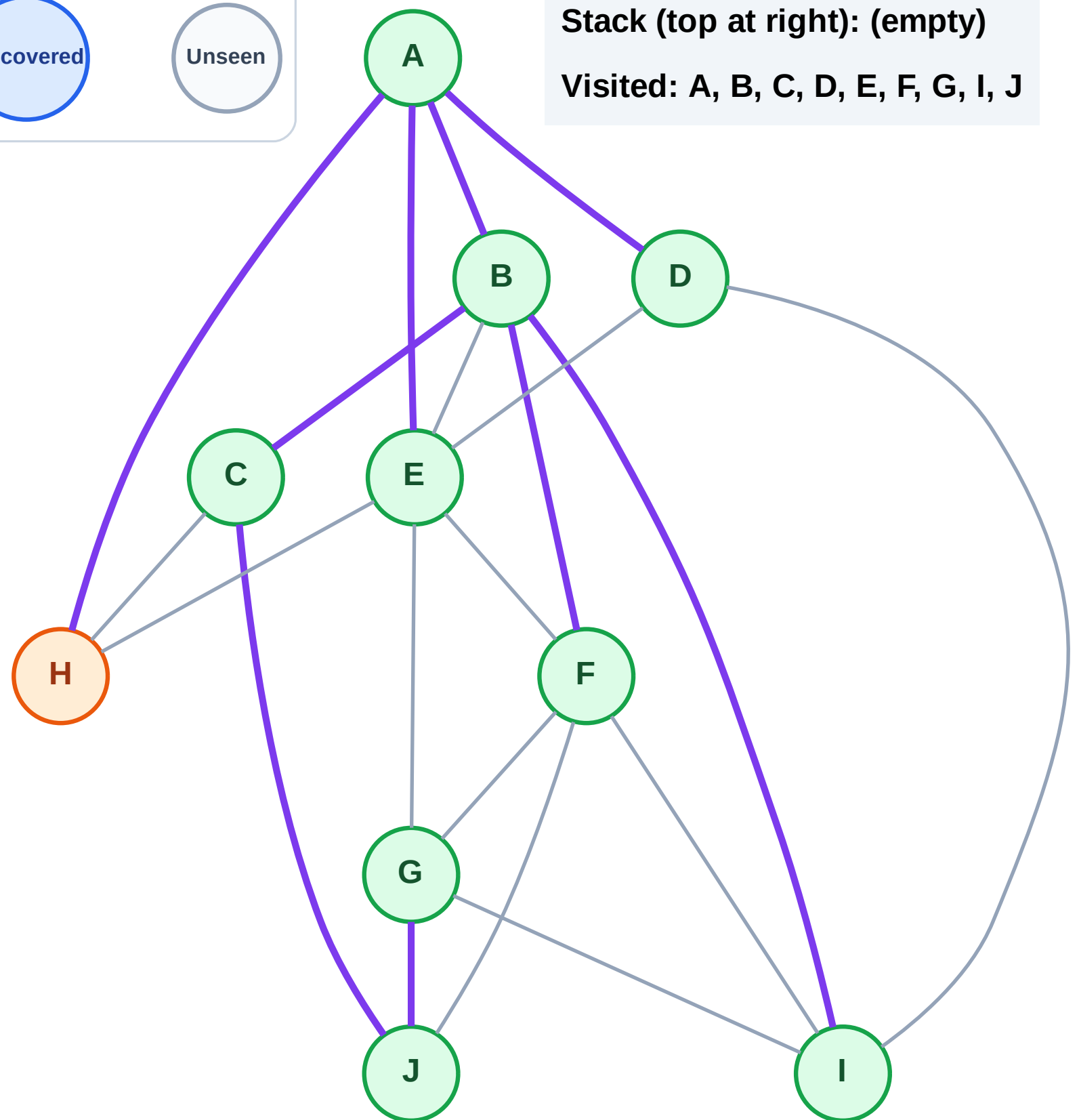
Step 20: Process node H

Legend



Stack (top at right): (empty)

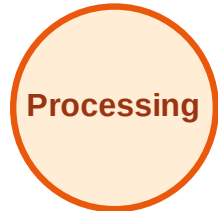
Visited: A, B, C, D, E, F, G, I, J



DFS Traversal

Step 21: No new neighbors from H

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

